



Microstructure-driven tunability of metamagnetic transitions in an all-d-metal Heusler shape memory alloy via simple heat treatments

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ABSTRACT

Metamagnetic shape memory alloys (MMSMAs) exhibit coupled magnetostuctural transitions that enable substantial changes in magnetic and mechanical properties triggered by martensitic transformations (MTs). Recently developed all-d-metal, off-stoichiometric NiCoMnTi Heusler alloys exhibit these functionalities at a lower cost than Ga-/In-containing MMSMAs. Here, we present a systematic study demonstrating process-only control of both the MT temperatures and the austenite Curie temperature (T_C) in all-d-metal MMSMAs at a fixed nominal composition. Simple, scalable two-step heat treatments allow process-only control of D0₂₄ (Ni,Co)₃(Ti,Mn) hexagonal precipitates, as shown by microstructural and compositional analyses. This study offers the first systematic evidence of how these precipitates evolve under various heat-treatment conditions and, for the first time in this system, uncovers their epitaxial relationship with the B2 austenitic matrix. Beyond these experimental observations, we provide a mechanistic understanding that links precipitate evolution to subtle changes in matrix stoichiometry and quantitatively connects these changes to variations in MT temperatures and T_C . The resulting microstructure-composition-property relationship explains how secondary heat treatments adjust the electron per atom ratio (e/a) and magnetic exchange to tune structural and magnetic transitions. These findings establish quantitative relationships between processing, microstructure, and magnetostuctural response, defining a processing-based pathway to tune functional performance in all-d-metal Ni-Co-Mn-Ti metamagnetic shape memory alloys.

1. Introduction

Metamagnetic shape memory alloys (MMSMAs) are characterized by their ability to undergo a martensitic transformation (MT), which modifies both the crystalline structure and magnetic behavior of these materials, turning the MT into a first-order magneto-structural transition [1–3]. The MT-induced structural change affects the magnetic state of the materials since the high temperature, high symmetry phase (austenite) and the low temperature, and in general lower symmetry phase (martensite) possess different magnetic states; paramagnetic (PM), ferromagnetic (FM), or antiferromagnetic (AFM) states can exist in either phase [4,5]. Consequently, in MMSMAs, the MT is sensitive not only to temperature and applied stress, as typically found in conventional shape memory alloys (SMAs), but also to applied magnetic fields [6–9]. Therefore, it is possible to purposefully manipulate these conditions to achieve distinct magnetic states, enabling large magnetization

changes through the MT.

In Ni–Mn-based MMSMAs, the magnetic behavior of these materials is predominantly governed by Ruderman-Kittel-Kasuya-Yosida (RKKY) exchange interactions [10,11]. Consequently, the strength of ferromagnetic and antiferromagnetic exchange interactions between the different atomic species, which determines the overall magnetic behavior of the material, is strongly influenced by the interatomic distance of Mn–Mn pairs [11–13]. These interatomic distances are significantly altered by the structural transition from austenite, typically an ordered body-centered cubic (BCC) phase for Ni–Mn-based alloys, to martensite, which can present a wider variety of structures, such as tetragonal, orthorhombic or monoclinic phases. This change often results in austenite being ferromagnetic (FM) while martensite may exhibit paramagnetism or weak antiferromagnetism (AFM) due to competition between FM and AFM interactions.

Notably, Ni–Mn-based alloys with the ternary element of p-group

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atoms, Ni-(Co)-Mn-X (X = In [14], Ga [15,16], Sn [17], Al [18]), can have different distributions of their atomic species and crystalline ordering depending on their composition and thermal processing, and may transition between two of these ordering states through an order-disorder (ODO) transition [19]. Among the different possible ordering configurations that 4 atomic species with pseudo-Heusler stoichiometry can adopt over the atomic packing of a BCC ($Im\bar{3}m$, W-like) structure [20,21], B2 ($Pm\bar{3}m$, CsCl-like), and $L2_1$ ($Fm\bar{3}m$, Cu_2MnAl -like) structures are often demonstrated by Ni-(Co)-Mn-X alloys for the compositional range for which MT takes place. For instance, previous works have demonstrated that alloys with X = Sn, Sb solidify to $L2_1$ [22], while alloys with X = Ga, In, or Al solidify to B2 phase but undergo a B2- $L2_1$ ODO transition (partial/weak for the Al case) at lower temperatures [23–25]. The structure of the B2 phase can be described using the same base cell as the BCC/A2 phase, but it requires differentiated atomic sites defined as the “center” and the “corners” of the cell, occupied by different atomic species. In Ni-Co-Mn-X alloys, these atomic sites are expected to be preferentially occupied by Ni/Co and Mn/X atoms, respectively [26,27]. By contrast, $L2_1$ requires a base cell called a superlattice, equivalent to 8 BCC cells, with at least 3 atomic sites to describe its chemical order, as the Mn/X atomic site in B2 splits into 2 distinguishable sites alternating Mn and X atoms along the $\langle 100 \rangle$ directions. These ordering states are characterized by different distributions of Mn atoms, affecting Mn-Mn distances, which then influence the strength of ferromagnetic and antiferromagnetic interactions via the RKKY principle. Consequently, the magnetic properties of MMSMAs, including Curie temperature (T_C) or magnetization changes associated with MT, depend on their degree of atomic order.

However, the brittle nature of Ni-Co-Mn-X alloys and the high cost of p-metal elements such as Ga and In make them impractical for certain applications. Since the covalency between p- and d-orbitals, so-called p-d orbital hybridization, has been suggested as responsible for their brittle nature [28], recent works have proposed addressing this challenge by replacing the p-metal element with another transition d-metal. In particular, the Ni-(Co)-Mn-Ti alloy system has attracted significant attention as a promising all-d-metal MMSMA, since the mechanical properties of Ni-(Co)-Mn-Ti alloys are reported to surpass those of conventional Ni-(Co)-Mn-X alloys, possibly due to improved phase stability [29] via d-d hybridization through covalent bonding [30–32]. Theoretical calculations by Yan et al. [30] show that Ni_2MnTi has more ductility than conventional MMSMAs, such as off-stoichiometric Ni_2MnGa , Ni_2MnIn , and Ni_2MnAl Heusler alloys, due to the absence of p-d orbital hybridization. In addition, recent reports have demonstrated that the multicaloric effects associated with the MT in Ni-Co-Mn-Ti alloys are comparable to those of p-metal Ni-Co-Mn-In alloys. For instance, $Ni_{1.42}Co_{0.58}Mn_{1.4}Ti_{0.6}$ exhibits an adiabatic temperature change of 11.5 K after the adiabatic application of a compressive stress of 175 MPa [33] and a barocaloric effect of -24.2 J/kg·K with an adiabatic temperature change of 4.2 K under a pressure of 1 kbar associated with an increase of MT temperatures of about 5 K [34]. Regarding the magnetocaloric effect, it is reported that $Ni_{37}Co_{13}Mn_{34}Ti_{16}$ showed 38 J/kg·K under a 2 T magnetic field with adiabatic temperature change of 3.5 K at around 223 K [35] and $Ni_{36}Co_{14}Mn_{35}Ti_{15}$ showed 19.3 J/kg·K under 5 T field at 264 K [31]. These promising caloric responses from Ni-Co-Mn-Ti MMSMAs, combined with their improved mechanical properties, enable their potential application as room-temperature (RT) solid-state refrigerants.

Co addition in Ni-Mn-based alloys has been shown to improve some of their functional properties [36–40]. Typically, Co atoms substitute Ni atoms and are expected to occupy the same atomic sites. As Co establishes stronger ferromagnetic interactions with the other atomic species compared to Ni [41], this substitution results in increased T_C and magnetization, especially for the cubic austenite phase. Consequently, Co substitution can enhance magnetization change during the MT, which can improve the low-field magnetocaloric response; however, at

higher magnetic fields, this benefit may be offset by a reduction of the total transition entropy change, as the positive lattice entropy contribution is increasingly compensated by a growing negative magnetic entropy contribution, thereby limiting the maximum achievable magnetocaloric effect [28,42]. In Ni-Co-Mn-Ti alloys, substituting 12–17 at.% Ni with Co may be necessary for the alloy to exhibit a transformation from FM austenite to PM martensite (depending on Ti content). The exact ratio of Ni/Co can be precisely selected to tune the magnetic martensitic phase transformation characteristics [28,35,43]. Historically, less than 10 at.% Co has been sufficient in most other conventional Heusler-type MMSMAs, promising for room temperature applications (e.g., $Ni_{45}Co_5Mn_{36.7}In_{13.3}$ [19] and $Ni_{45}Co_5Mn_{33}Sn_{17}$ [38]).

In addition to compositional changes, recent work has shown that the degree of B2 atomic ordering can be modified by high-temperature thermal processing (e.g., 973–1173 K) in $Ni_{37}Co_{13}Mn_{33}Ti_{17}$, affecting its MT behavior [43]. However, as high atomic diffusion rates are expected in this temperature range, these treatments may also induce other microstructural changes that potentially influence MT behavior in various SMA systems, commonly resulting in enhanced homogeneity or precipitation. For instance, in binary Ni-Ti and ternary Ni-Ti-Hf/Zr alloys [44–46], MT characteristics have been controlled by precipitate formation and growth. In Ni-Co-Mn-X alloys, thermal processing may alter MT characteristics by modifying the degree of atomic ordering. For example, Ni-Co-Mn-In alloys exhibit strong correlations between MT characteristics and features of the atomic order microstructure, such as the volume fraction of B2/ $L2_1$ phases or the average $L2_1$ domain size [47–49]. Conversely, Ni-Co-Mn-Sn alloys show only minimal changes in ordering depending on thermal processing conditions [17].

Beyond atomic ordering, microstructural processes such as precipitation phenomena during heat treatment can significantly influence the transformation and magnetic behavior of Ni-Mn-based Heusler alloys. In Ni-Mn-X (X = In, Sn, Sb) alloys, aging can induce compositional segregation and the formation of Ni-rich precipitates, typically Ni_2MnX , that modify the matrix composition and magnetic interactions. However, their formation is often promoted under an applied magnetic field, which stabilizes ferromagnetic regions within an antiferromagnetic matrix [50–52]. In contrast, Ni-Mn-Ga alloys, particularly in Mn-rich compositions, undergo thermally driven γ precipitation during thermal aging, regardless of the presence of a magnetic field [53]. These Mn-rich secondary phases refine the austenitic matrix, introduce compositional modulation, and influence MT properties. Although such precipitation mechanisms are well established in conventional Ni-Mn-based systems, it remains unclear whether similar precipitation behavior occurs in all-d-metal Ni-Co-Mn-Ti alloys. Understanding how different thermal processing routes affect possible precipitation, ordering, and microstructural evolution in this system is therefore critical for revealing the origins of its martensitic and magnetic tunability.

Recent studies on thermal processing of Ni-Co-Mn-Ti alloys have primarily focused on optimizing solution heat treatment (SHT) conditions [35] or characterizing the degree of B2 ordering at selected temperatures [43]. Koch et al. [43] noted the presence of precipitates in $Ni_{37}Co_{13}Mn_{33}Ti_{17}$ after annealing at 973 K and 1173 K, although no systematic variation of aging duration or chemical characterization of the precipitates was reported. In ternary Ni-Mn-Ti alloys, Yan et al. [30] and Villa et al. [54] reported the formation of (Ni,Mn) $_3$ Ti precipitates during thermal processing. However, a systematic investigation connecting secondary heat treatment (SecHT) temperature and duration to precipitate evolution, matrix composition changes, and the resulting MT and magnetic properties at a fixed nominal composition has not been reported for quaternary Ni-Co-Mn-Ti alloys. In particular, whether the changes in MT behavior induced by SecHT are primarily due to variations in atomic ordering, as observed in Ni-Co-Mn-X (X = In, Ga, Sn) MMSMAs [47–49], precipitate formation and growth as in Ni-Ti and Ni-Ti-Hf/Zr alloys [44–46], or other microstructural changes such as enhanced chemical homogeneity, remains an open question.

To address this question, the present study investigates the

microstructural and compositional modifications induced by secondary heat treatments in Ni–Co–Mn–Ti MMSMAs. Specifically, SecHT conditions are systematically varied across four temperatures (873 K to 1173 K) and multiple durations at fixed nominal composition, enabling quantitative correlations between precipitate area fraction, matrix composition, martensite start temperature (M_S), and austenite Curie temperature (T_C). The study reveals that $\text{DO}_{24}(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitation is the dominant mechanism controlling the SecHT-dependent MT behavior, and identifies the epitaxial orientation relationship between these precipitates and the B2 austenitic matrix for the first time in this alloy system. A quantitative kinetic framework based on the Johnson-Mehl-Avrami-Kolmogorov model and Arrhenius analysis is developed, providing a precipitation solvus estimate, an apparent activation energy, and a time-temperature-transformation diagram that offers practical guidance for heat treatment design. The resulting insights define processing windows for tailoring transformation characteristics and magnetic behavior in device-relevant applications such as solid-state refrigeration and actuation.

2. Experimental procedure

Polycrystalline Ni–Co–Mn–Ti alloys with the nominal compositions of $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{Ti}_{15}$, where $x = 13.5, 14.0, 14.5$ and 15.0 (all compositions in at.%), were prepared as arc-melted buttons of approximately 10 g each in an ultra-high purity argon (99.999% purity) environment using high-purity Ni and Ti (99.98%), Mn and Co (99.95%) elements. In the following, these compositions will be denoted as Co13.5, Co14, Co14.5 and Co15, respectively. After the initial melting and cooling, the samples were flipped and remelted 4 more times to enhance chemical homogeneity. To compensate for Mn evaporation losses during melting, 1.5 wt.% excess Mn was added to the weighed elemental mixture for each alloy composition. The arc-melted samples were then sealed in Quartz Tubes (QTs) under ultra-high purity argon, solution heat treated (SHTed) at either 1173 K for 24 h or 1323 K for 96 h [35] and subsequently water quenched (WQ) to room temperature (RT) for comparison of chemical homogeneity. To investigate the effect of different SHT conditions, Co14.5 samples were analyzed using wavelength-dispersive spectroscopy (WDS) on a Cameca SXFive electron probe microanalyzer (EPMA), and their thermomagnetic and calorimetric responses were measured using a TA Instruments DSC Q2000 and a Quantum Design MPMS SQUID-VSM magnetometer.

Multiple specimens with a mass of 100–200 mg were sampled from each arc-melted $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{Ti}_{15}$ alloy ($x = 13.5, 14, 14.5$, and 15 at.%) for calorimetric measurements, and a portion of these specimens, approximately 30 mg, was used for magnetic measurements. To obtain calorimetric response during MT, a TA Instruments DSC Q2000 was used to identify MT temperatures and the enthalpy change associated with this phase transition. The temperature range spanned from 223 K to 423 K to fully capture the whole MT region with a heating/cooling rate of 10 K/min. A Quantum Design MPMS SQUID-VSM magnetometer was employed to measure the thermomagnetic (M-T) and field-induced magnetic (M-H) responses of the $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{Ti}_{15}$ samples with varying Co content. Temperature-dependent measurements were performed under constant applied magnetic fields of 0.05, 1, and 7 T, over a range from 150 K to 400 K, using a heating/cooling rate of 5 K/min. Isothermal magnetic cycles were carried out over a broad temperature range, from 400 K down to either 50 K or 10 K, spanning the martensite-to-austenite transition. These measurements captured the magnetic field dependence (0–7 T) of the martensite and austenite phases (outside the MT region), as well as the mixed-phase states during partial or full magnetic field-induced MT, depending on temperature. The resulting M-H curves were used to estimate the magnetocaloric effect (MCE) using the Maxwell relation [55]:

$$\Delta S_T(T, H) = \mu_0 \int_{H_1}^{H_2} \left(\frac{\partial M}{\partial T} \right)_H dH \quad (1)$$

where ΔS_T is the isothermal entropy change resulting from changing the applied magnetic field from H_1 to H_2 at a given temperature T . Before each M-H experiment, the sample was cooled below martensite finish (M_F) temperature and then heated back to the target temperature. As a result, the estimated ΔS_T primarily corresponds to the MCE associated with the induction of the martensite-to-austenite transition when the transformation occurs between the austenite start (A_S) and austenite finish (A_F) temperatures, i.e., the MCE during “heating”. Note that, due to thermal hysteresis, the initial volume fraction of austenite/martensite at a given temperature depends on the thermal history; thus, one can distinguish between MCE during “cooling” and “heating” states.

After testing the lab-scale arc-melted samples, a cylindrical bar (38 mm in diameter, 255 mm in length) of nominal $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ at.% (Co13.5) was fabricated by Sophisticated Alloys, Inc. using vacuum induction melting and used to study the effects of SecHTs on the MT and magnetic characteristics of Ni–Co–Mn–Ti MMSMAs. Multiple 1-mm thick disk specimens with a diameter of 4.5 mm were cut using wire electrical discharge machining. The specimens were sealed in QT under an argon atmosphere, SHTed at 1323 K for 96 h, and then WQ to RT. Subsequently, each specimen was SecHTed at 873 K, 973 K, 1073 K, or 1173 K for durations of 20 min, 1 h, 3 h, 24 h, or 72 h, creating a systematic set of thermal processing conditions. In addition, due to the observed change in the trend of onset temperature of MT (martensite start, M_S) after 24 h of SecHT at 873 K, further SecHTs of 12 h and 36 h were specifically carried out at this temperature. The MT behavior of these samples was evaluated by both calorimetry and magnetometry under the same experimental conditions described above. An FEI Quanta 600 FE-SEM scanning electron microscope (SEM) was employed to quantitatively assess both the area fraction of the secondary phases and the matrix composition. Precipitate area fractions were determined by averaging measurements from ten distinct regions per SEM sample, with all image analyses performed using ImageJ software. Chemical compositions of precipitates and matrix were determined using Wavelength Dispersive Spectroscopy (WDS). Data were collected using a LaB6-equipped CAMECA SXFive electron probe microanalyzer at 15 kV and 20 nA with a pixel dwell time of 115 μs and a step size of 0.7 μm .

The crystalline structures of both the precipitate and matrix phases were first examined by X-ray diffraction (XRD) on Co13.5 samples that had been SHTed at 1323 K for 96 h and then subjected to SecHT at 973 K for 3 h, 24 h, and 72 h, using a Bruker D8 Discover system with a $\text{Cu K}\alpha$ source. For transmission electron microscopy (TEM) analysis, including energy-dispersive X-ray spectroscopy (EDS), Co13.5 specimens SHTed at 1323 K for 96 h followed by SecHT at 873 K for 24 h and 1173 K for 1 h, as well as a Co14.5 sample SHTed at 1323 K for 96 h were selected. After the specified heat treatments, ~ 100 μm -thick sheets were punched into 3 mm discs and electropolished in 30 vol.% nitric acid in methanol at approximately 253 K. Final imaging, structural, and compositional analysis were carried out on a Thermo Fisher Talos F200i Scanning TEM (STEM) equipped with an EDS detector.

3. Experimental results

3.1. Optimal solution heat treatment condition and chemical homogeneity

The chemical homogeneity of the Co14.5 alloy was evaluated before (as-cast) and after two different SHT conditions, at 1173 K for 24 h and at 1323 K for 96 h, for comparison. The SEM backscattered electron (BSE) image in Fig. 1(a) demonstrates that the as-cast condition exhibits dendrites (light gray) with thicknesses of a few microns, which are Mn/Ni-rich and Ti/Co-lean relative to the nominal composition, consistent with previous EDS results on as-cast Ni–Co–Mn–Ti alloys [33,35]. The greatest compositional variability is observed in Mn, with differences of up to 10 at.% between dendritic and interdendritic regions, while the maximum chemical contrasts for Ti, Co, and Ni are approximately 7, 3, and 2 at.%, respectively. Additionally, a clear elemental partitioning is

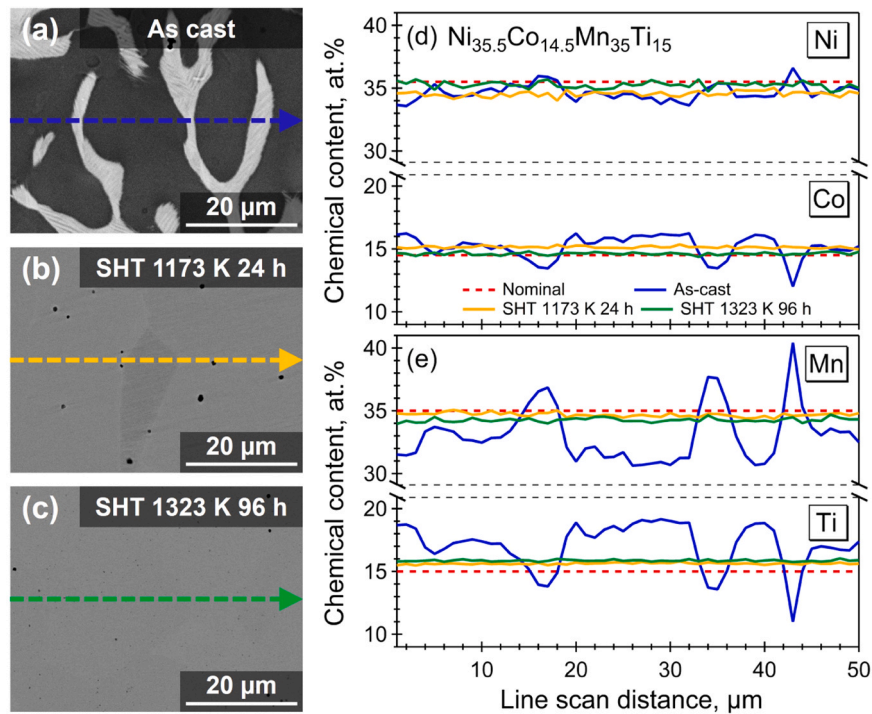


Fig. 1. SEM BSE micrographs showing the microstructure of $\text{Ni}_{35.5}\text{Co}_{14.5}\text{Mn}_{35}\text{Ti}_{15}$ MMSMA samples in (a) as-cast condition and after SHT at (b) 1173 K for 24 h and (c) 1323 K for 96 h. Chemical compositions measured by WDS along the line scans, indicated by arrows in (a), (b), and (c), show the distribution for (d) nickel (Ni) and cobalt (Co) and (e) manganese (Mn) and titanium (Ti), for the as-cast (solid blue), 1173 K for 24 h (solid orange), and 1323 K for 96 h (solid green) conditions. In panels (d) and (e), the measured elemental compositions are also compared to the nominal target composition (dashed red line).

observed between dendritic and interdendritic regions: the combined Mn+Ti and Ni+Co contents remain more stable than the variation of individual elements. The darker interdendritic regions have a composition close to $(\text{NiCo})_{50}(\text{MnTi})_{50}$ at.%, while the dendrites are closer to $(\text{NiCo})_{48}(\text{MnTi})_{52}$ at.%. In contrast, Figs. 1(b) and 1(c) show that SHT at 1173 K and 1323 K, respectively, effectively homogenizes the alloy, eliminating the dendritic structure. Only a weak contrast remains, corresponding to grains with different crystallographic orientation.

The variation in chemical composition along a line scan for samples in the as-cast condition and after SHT at 1173 K or 1323 K is shown in Figs. 1(d) and 1(e). These plots confirm that the compositional differences characteristic of the dendritic microstructure are no longer present in the homogenized samples, i.e., local chemical variations after both heat treatments are minor, considering the precision of the WDS technique.

Fig. 2 presents the temperature dependence of the heat flow (Fig. 2(a)), and the magnetization (Fig. 2(b)), during different thermal cycles

for $\text{Co}_{14.5}$ samples subjected to two different SHT conditions: 1173 K for 24 h and 1323 K for 96 h. Both calorimetric and thermomagnetic measurements point out that the sample SHTed at 1173 K exhibits a broader MT range, e.g., M_S-M_F (M_F , martensite finish temperature), than the sample SHTed at 1323 K. This broader MT range has been attributed to poorer chemical homogeneity [35], which cannot be detected with the resolution of the WDS technique. Furthermore, 1323 K/96 h case shows a smaller transformation hysteresis (A_P-M_P , where A_P and M_P are austenite and martensite transformation peak temperatures) and more pronounced exothermic (on cooling) and endothermic (on heating) peaks. The DSC-integrated total enthalpy is 5.2 J/g for the 1173 K/24 h and 7.2 J/g for the 1323 K/96 h. The observed difference can be attributed to transition smearing and retained austenite, as indicated by the broader martensitic transformation and increased low-temperature magnetization in the 1173 K/24 h sample shown in Fig. 2(b), rather than a change in intrinsic latent heat. A broader transformation distributes latent heat over a wider temperature range, resulting in a

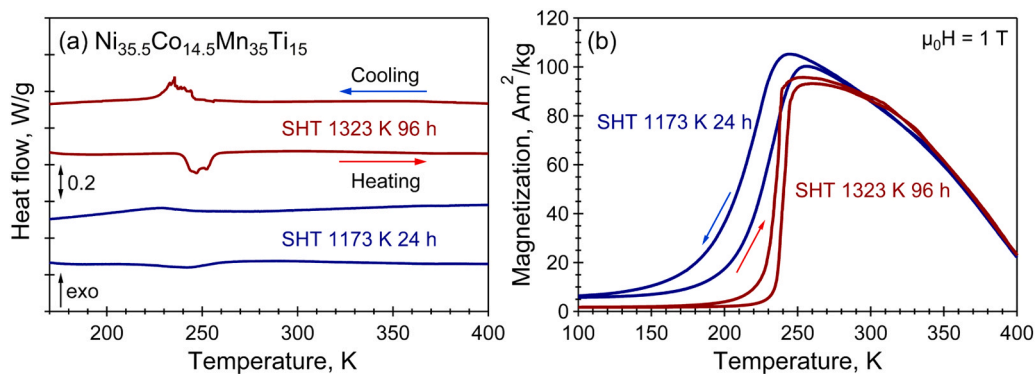


Fig. 2. Transformation characteristics of two $\text{Ni}_{35.5}\text{Co}_{14.5}\text{Mn}_{35}\text{Ti}_{15}$ MMSMA samples subject to SHT conditions at either 1323 K for 96 h (red) or 1173 K for 24 h (blue) shown by (a) calorimetric response and (b) thermomagnetic response under 1 T magnetic field.

portion of the signal falling outside the practical DSC integration window. These factors decrease the measured peak area and the apparent integrated enthalpy change, although the intrinsic latent heat per fully transformed mass remains comparable. Despite the very similar chemical profiles obtained by WDS, the MT characteristics of the 1323 K/96 h sample suggest that this treatment results in a higher degree of chemical homogeneity compared to the 1173 K/24 h condition. This result is consistent with the findings of Taubel et al. [56], which showed that SHT at 1323 K for 96 h results in a chemically homogeneous microstructure and a correspondingly sharper martensitic transformation characterized by a reduced transformation range. Consequently, all subsequent SHTs described in this study were conducted at 1323 K, as this condition is considered as optimal (as compared to the SHTs conducted at 1173 K) for the SHT of $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{Ti}_{15}$ alloys, $x = 13.5, 14.0, 14.5$ and 15.0 at.%.

3.2. The effect of cobalt substitution on the martensitic transformation

Fig. 3(a) presents the magnetic response of Co13.5, Co14, Co14.5, and Co15 alloys under SHT conditions during a thermal cycle performed under a constant applied magnetic field of 1 T. Table 1 shows the Ni/Co content dependence of the MT temperatures and enthalpy, entropy, magnetization changes, thermal hysteresis, and the Curie temperature (T_C) of these alloys. As shown in Fig. 3(a), increasing Co content correlates well with a decrease in the M_S temperature and an increase in T_C . The inverse correlation between M_S and T_C with increasing Co content reflects the strengthening of ferromagnetic order in the austenite, arising from enhanced Mn-Co-Mn exchange interactions [28]. These interactions reduce the magnetic free energy and increase the thermodynamic stability of the ferromagnetic austenite relative to the paramagnetic martensitic phase. Comparable compositional dependence has been widely reported for Ni–Co–Mn–Ti Heusler alloys [28,31,42,57], confirming the well-established magnetic-structural coupling in this alloy system. For the present work, each additional percent of Co lowers M_S by ~ 65 K between 13.5 and 15 at.% Co for the 1173 K/24 h SHT case, while for the 1323 K/96 h one, M_S becomes more Co-sensitive with an ~ 80 K decrease for the same composition range. An additional result of increasing Co content from 13.5 to 15 at.% is an enhanced difference between the T_C of the austenite and M_S , $T_C - M_S$. This parameter has been used in earlier studies on Ni–Mn–X MMSMAs as an indicator of the relative stability of the ferromagnetic austenite compared to the weakly magnetic martensite, which can also be correlated with various MT characteristics, such as an increased thermal hysteresis or a larger magnetization change during MT, as seen in Table 1 [58].

These correlations arise primarily from the dependence of the total MT entropy change, $\Delta S_{\text{total}} = S_{\text{mar}} - S_{\text{aus}}$, on the ferromagnetic ordering of the austenite, as the magnetic contribution to the entropy change, ΔS_{mag} , is expected to be positive during the transition from FM austenite to PM

martensite [16,49,59]. Therefore, enhancing the ferromagnetic ordering of the austenite within the MT temperature range enhances ΔS_{mag} but decreases ΔS_{total} , thereby stabilizing the austenite phase relative to the weakly magnetic martensite phase. Increasing Co strengthens the ferromagnetic ordering of the austenite, which simultaneously stabilizes this phase relative to the paramagnetic martensite and reduces the total transition entropy, as the magnetic contribution compensates the nearly composition-independent structural entropy [57]. This behavior is reflected in a systematic decrease in the transformation enthalpy (Δh) and entropy (Δs) with increasing Co content. Specifically, Δh decreases from 13.5 J/g to 3.7 J/g and Δs from 46 J/kg·K to 17 J/kg·K as Co increases from 13.5 to 15 at.% (Table 1). Guan et al. [31] reported a similar trend, with Δh and Δs decreasing from 9.7 to 4.5 J/g and 34 to 19 J/kg·K, respectively, over the same Co range. However, the present values show slight deviations from those previously reported, which can be attributed to differences in alloy composition and SHT conditions.

Fig. 3(b) shows the temperature dependence of the isothermal entropy change (ΔS_T) resulting from the application of a magnetic field of either 2 T (accessible with permanent magnets) or 7 T (maximum field employed during the isothermal magnetic field dependent measurements) for the martensite-to-austenite transition in $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{Ti}_{15}$ with x ranging from 13.5 to 15.0 at.%. In both cases, the largest ΔS_T is observed in Co14.5, reaching approximately 15.5 J/kg·K at 240 K and 30 J/kg·K at 232 K for 2 and 7 T, respectively. Note that these values correspond to an initial magnetic field loading without considering the effect of hysteresis. Comparing the maximum isothermal entropy changes obtained for 7 T with the MT entropy changes derived from the DSC curves (summarized in Table 1), the ΔS_T values for Co13.5 and Co14 alloys are lower than expected from DSC, 26 versus 47 J/kg·K and 27 versus 36 J/kg·K, respectively. In addition, if a Clausius-Clapeyron type relationship, $\Delta s \approx \Delta M \left(\frac{\Delta H_C}{\Delta T} \right)$, is used to estimate entropy change from the shift of the critical field, H_C , needed to induce the transformation with temperature values of approximately 40 and 35 J/kg·K are obtained, respectively, which are closer to the calorimetry-based estimates. The main reason behind this discrepancy is the reduced sensitivity of the critical transformation temperatures (A_S in the case of the induced reverse MT) to the magnetic field, resulting from the decreasing MT magnetization difference, ΔM , and increasing MT entropy difference, Δs , when Co content decreases from 15 to 13.5 at.%. In this case, measured temperature shifts upon the application of the magnetic field are approximately 1.5, 2, 3.5, and 5.5 K/T for Co13.5, Co14, Co14.5, and Co15, respectively. Consequently, a field of 7 T produces a shift of around or below 15 K for Co13.5 and Co14, insufficient to induce the full MT at any given temperature since the MT range, $A_F - A_S$, is about 20 K for both alloys.

Based on the calorimetric and thermomagnetic results for MT temperatures and the magnetocaloric effect (MCE) in the four lab-scale arc-melted samples, $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ (Co13.5) has been considered for

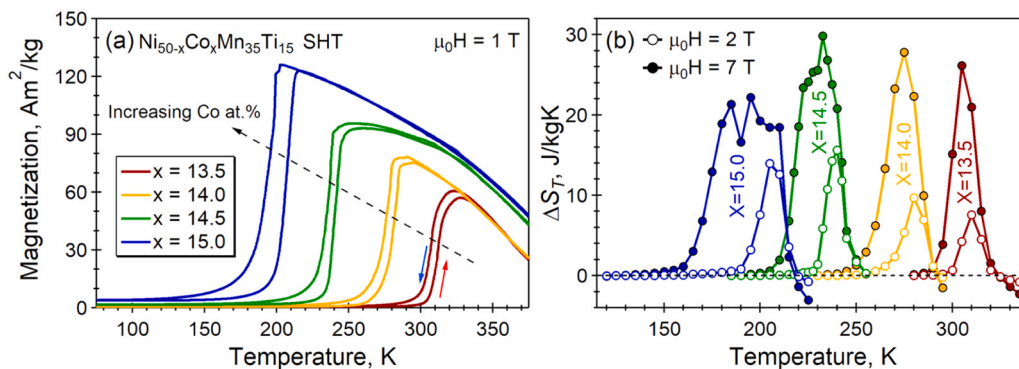


Fig. 3. Temperature dependence of (a) magnetization under a constant applied magnetic field of 1 T and of (b) isothermal entropy change, ΔS_T , after the application of a magnetic field of either 2 T (open symbols) or 7 T (full symbols) as calculated from magnetic field dependent magnetization measurements for $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{Ti}_{15}$ alloys with x varying Co content from 13.5 to 15.0 at.%.

Table 1

Properties of $\text{Ni}_{50-x}\text{Co}_x\text{Mn}_{35}\text{Ti}_{15}$ were obtained from thermomagnetic and calorimetric measurements. Enthalpy (Δh) and entropy (Δs) changes were obtained as $\int \delta q$ and $\int \delta q/T$ using calorimetric measurements (DSC) at a heating/cooling rate of 10 K/min, respectively, where δq is the instantaneous change in the heat produced by the MT. Magnetization change (ΔM) was measured under a magnetic field of 1 T during cooling at a rate of 5 K/min. Transformation temperatures were measured in a magnetic field of 0.01 T using the intersection tangent lines method.

Sample description		Properties from cooling (10 K/min)				Properties from heating (10 K/min)				Thermal hysteresis	Curie temperature	Magnetization change
Alloy #	Co at. %	M_S K	M_F K	Δh J/g	Δs J/kg·K	A_S K	A_F K	Δh J/g	Δs J/kg·K	$A_F - M_S$ K	T_C K	ΔM Am ² /kg
1	13.5	324.3	292.6	13.5	46.1	305.8	326.9	-14.6	-48.3	2.6	348.4	56.6
2	14.0	288.4	261.5	11.5	39.4	266.9	293	-10.4	-34.7	4.6	351.3	74.8
3	14.5	244.9	230.5	7.2	30.4	237.7	252.1	-7.8	-31.5	7.2	376.2	91.5
4	15.0	203.4	170.8	3.7	16.9	193.5	215.4	-3.3	-14.6	12	379.1	119.1

further analysis of transformation tailorability. This composition was chosen due to its favorable MT temperatures near room temperature, low thermal hysteresis and its reasonable MCE, despite its full possible Δs_T was not achieved with a field of 7 T. Overall, Co13.5 is a promising candidate for solid-state cooling applications.

3.3. Effects of secondary heat treatments

3.3.1. Martensitic transformation and magnetic properties

The thermomagnetic responses of Co13.5 samples subjected to the different thermal processing conditions (SecHTs) are depicted in Fig. 4 (a–d) for thermal cycles performed under a constant magnetic field of 1 T. The main trend observed in Fig. 4 is that increasing SecHT duration results in a significant shift of the M_S to lower values across all studied SecHT temperature conditions. Furthermore, this shift in the MT temperatures is accompanied by increases in thermal hysteresis, transformation range, and magnetization change, the latter being related to the MT moving further away from the T_C temperature of austenite. These changes in MT characteristics are most pronounced for SecHTs performed at 873 K and 973 K, diminish with increasing SecHT temperature to 1073 K, and are nearly absent at 1173 K, which produces minimal changes in M_S and hysteresis as compared to the SHT condition (1323 K).

The thermomagnetic measurements in Fig. 4(a–b) also indicate that Co13.5 samples subjected to SecHT durations longer than 3 h at 873 K or 1 h at 973 K exhibit incomplete MTs, meaning that retained austenite exists at cryogenic temperatures, well below M_F . In contrast, this MT arrest behavior is not observed for the higher SecHT temperature cases in Fig. 4(c–d), which display very low magnetization at low temperatures. Notably, for both 873 K and 973 K cases, the low-temperature magnetization, and thus the amount of retained austenite, reaches a maximum for a SecHT duration of 24 h.

A summary of the M_S temperatures for different SecHT temperature and duration conditions is given in Fig. 5(a). Although the SecHT duration is plotted on a logarithmic scale, the leftmost point on the x-axis is defined as zero SecHT duration (i.e., the sample subjected only to SHT) for the sake of convenience. This reference point represents the initial state of the Co13.5 samples prior to any SecHT. As previously noted, Fig. 5(a) shows that increasing SecHT duration generally leads to a decrease in M_S temperature. However, this trend is not uniform and is strongly dependent on SecHT temperature, potentially exhibiting distinct stages of microstructural evolution. In particular, the Co13.5 samples heat treated at 873 K or 973 K display an initial stage in which M_S decreases gradually with increasing duration, up to 3 h and 1 h cases, respectively. This is followed by a second stage characterized by a more pronounced temperature drop, persisting up to 24 h. Finally, a third

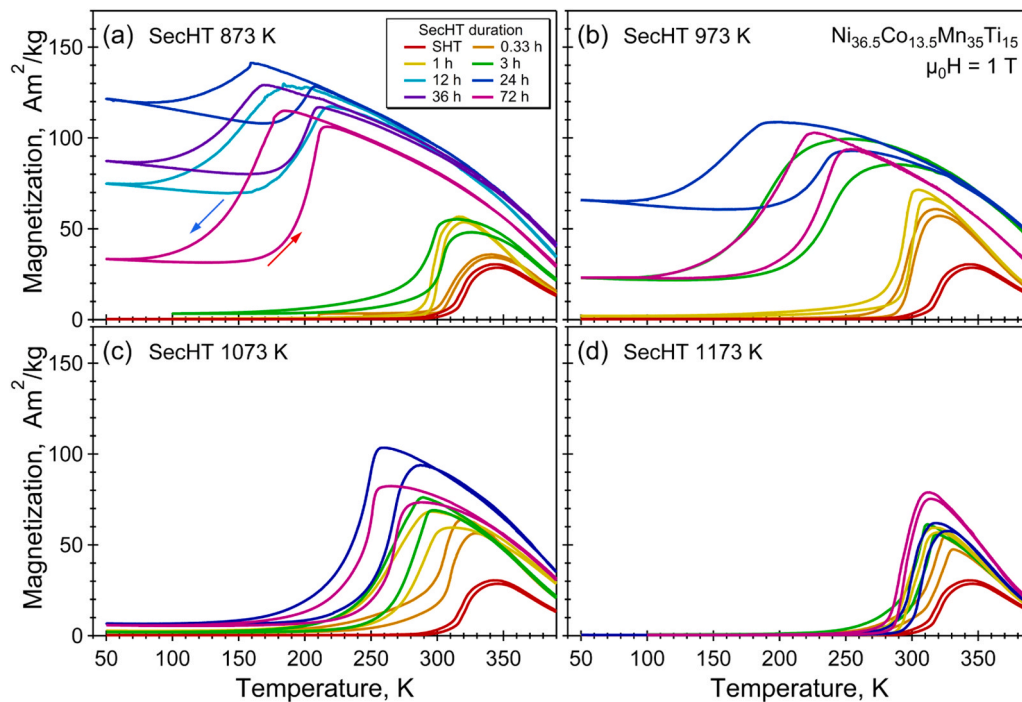


Fig. 4. Thermomagnetic response curves measured under a constant magnetic field of 1 T for $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ samples following high-temperature SecHTs. Panels (a) through (d) display data for SecHTs of 873 K, 973 K, 1073 K, and 1173 K, respectively. Each curve is color-coded to indicate a distinct SecHT duration.

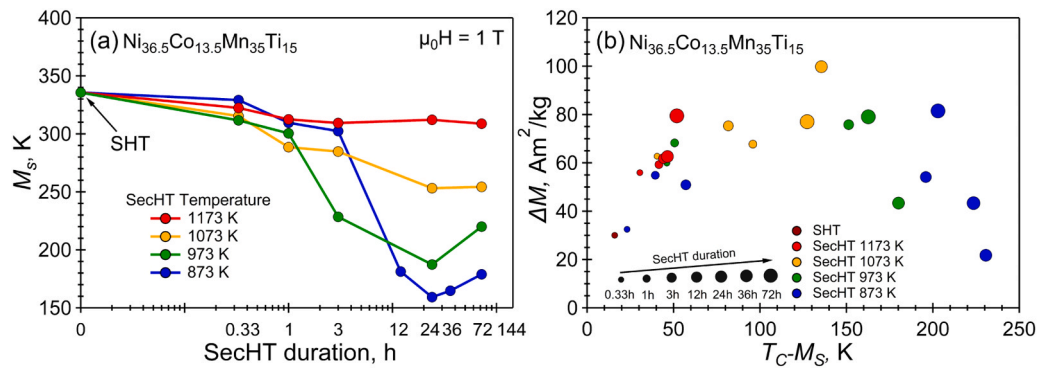


Fig. 5. (a) Summary of M_S transformation temperatures of $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ after high temperature SecHTs at 873 K, 973 K, 1073 K, and 1173 K for different durations under 1 T magnetic field. The starting point at “zero” SecHT duration represents the response of the initial SHT state. (b) Variation of the magnetization change, ΔM , as a function of $T_C - M_S$ for samples subjected to SecHTs. Symbol size reflects SecHT duration as indicated. Please note that 12 h and 36 h SecHT conditions exist only at 873 K.

stage shows a slight increase in M_S temperature between 24 h and 72 h. Consequently, both 873 K and 973 K conditions exhibit a minimum in M_S at 24 h. These trends are not observed in samples subjected to SecHT at 1073 K or 1173 K, which shows a more stable, logarithmic dependence of M_S on SecHT duration, with this effect diminishing at higher SecHT temperature.

Fig. 5(b) shows that the magnetization change (ΔM) varies non-monotonically with SecHT temperature and duration, closely related to the evolution of $T_C - M_S$. As $T_C - M_S$ increases from ~ 16 K in the SHT case to ~ 135 K after SecHT at 1073 K/24 h, ΔM increases from 30 Am²/kg to its maximum value of 100 Am²/kg. When $T_C - M_S$ exceeds about 135 K, however, the greater magnetic stability of the austenite partially suppresses the MT, retaining austenite at low temperatures and decreasing ΔM . This overall trend is reflected in the individual SecHT conditions. At 873 K, ΔM increases from 30 Am²/kg in the SHT state to 55 Am²/kg after 3 h as $T_C - M_S$ increases from 16 K to 60 K, then drops to 22 Am²/kg at 24 h when $T_C - M_S$ exceeds 200 K, and partially recovers to 81 Am²/kg at 72 h as $T_C - M_S$ decreases. At 973 K, a similar trend occurs with ΔM rising to 76 Am²/kg after 3 h, dropping to 43 Am²/kg at 24 h, and recovering to 79 Am²/kg at 72 h. At 1073 K and 1173 K, where $T_C - M_S$ remains below ~ 135 K throughout the entire SecHT duration range, the MT is complete and ΔM varies gradually with SecHT duration. The optimal balance between magnetic stabilization and transformation completeness, yielding the largest ΔM of 100 Am²/kg, occurs at 1073 K/24 h where $T_C - M_S$ is ~ 135 K.

The incomplete MT observed at long SecHT durations at 873 K and 973 K, where $T_C - M_S$ exceeds ~ 135 K, can be understood through the thermodynamic balance between structural and magnetic entropy contributions to the transformation. The total entropy change driving the MT (ΔS_{total}) consists of a structural/vibrational term (ΔS_{vib}) that favors the transformation and is approximately composition-independent (~ 65 J/(kg·K)) [57], and a magnetic term (ΔS_{mag}) that depends on the difference in magnetic ordering between the ferromagnetic austenite and the paramagnetic/weakly magnetic martensite. When M_S is close to T_C , the ferromagnetic ordering of austenite is weak and ΔS_{mag} is small, so ΔS_{total} is dominated by the structural term and the MT proceeds to completion. However, as M_S decreases progressively further below T_C with increasing SecHT duration, the austenite becomes strongly ferromagnetically ordered at the transformation temperature, and ΔS_{mag} grows and increasingly compensates ΔS_{vib} . When ΔS_{total} decreases significantly, the thermodynamic driving force for the MT weakens and the transformation becomes increasingly incomplete, leaving retained austenite at cryogenic temperatures. This entropy compensation mechanism is consistent with similar observations in other Ni–Mn-based MMSMAs [14,16,58,60], and the incomplete MT is observed specifically in the SecHT conditions that produce the largest $T_C - M_S$ separation.

3.3.2. Microstructure

SEM-BSE images in Fig. 6 compare the microstructure of SHT condition (Fig. 6(a)) with those of the samples heat-treated at 1073 K for different durations (Fig. 6(b–f)). These micrographs reveal the presence of a second phase in the form of microscale precipitates after all SecHTs at 1073 K, which are absent in the initial SHT state. Each specimen also contains pores, but the area fraction of these pores remains essentially unchanged across the heat treatment conditions examined.

As the SecHT duration increases from 0.33 h to 24 h, both the size and density of the precipitates increase. At 0.33 h, thin, needle-shaped precipitates (< 3 μm long) begin to form and are sparsely distributed. By 1 h, the number density increases markedly, though their morphology remains similar. At 3 h, the precipitate area fraction increases, with a higher population of precipitates measuring approximately 2–3 μm in length. Aging for 24 h yields the highest area fraction, with precipitates growing to about 4–5 μm in length, thickening, and achieving a more uniform distribution within the matrix. Finally, after 72 h, Ostwald ripening [61,62] becomes apparent: small needle-shaped precipitates dissolve while larger ones grow beyond 5 μm and widen, leading to a slight drop in number density and a broader size distribution. These observations suggest that a 24 h SecHT at 1073 K optimizes precipitate volume fraction and uniformity, whereas extended ageing promotes coarsening.

The micrographs in Fig. 7 compare the microstructures of samples treated for 24 h at different temperatures, corresponding to the conditions that resulted in the largest M_S shifts relative to the SHT case. Again, the observed microstructures confirm that second phases are formed after all SecHT conditions. In addition, there are significant differences in volume fraction, size, and interparticle spacing of these precipitates depending on the SecHT temperature.

Precipitate size increases with SecHT temperature from the 873 K/24 h case (Fig. 7(a)) to the 1073 K/24 h case (Fig. 7(c)), while the 973 K/24 h condition appears to yield the highest precipitate volume fraction. However, the volume fraction in the 873 K/24 h sample is difficult to quantify by SEM/BSE due to the small size of the fine precipitates. Note that intragranular precipitates for the 873 K/24 h case are hardly visible without the additional magnification of the inset in Fig. 7(a). The observed increase in precipitate size with SecHT temperature is consistent with the kinetics of precipitation, assuming the solid solution state occurs at temperatures above 1073 K. At 873 K, the larger undercooling (ΔT)—the difference between the heat treatment and precipitation solvus temperature—provides a higher driving force for nucleation, resulting in more nucleation sites but limited coarsening due to the low temperature, thus, yielding smaller precipitates. In contrast, at 1073 K, fewer nucleation sites exist due to lower ΔT , but the higher temperature promotes greater precipitate growth. The 1173 K/24 h treatment (Fig. 7(d)) results in a significant reduction in both precipitate

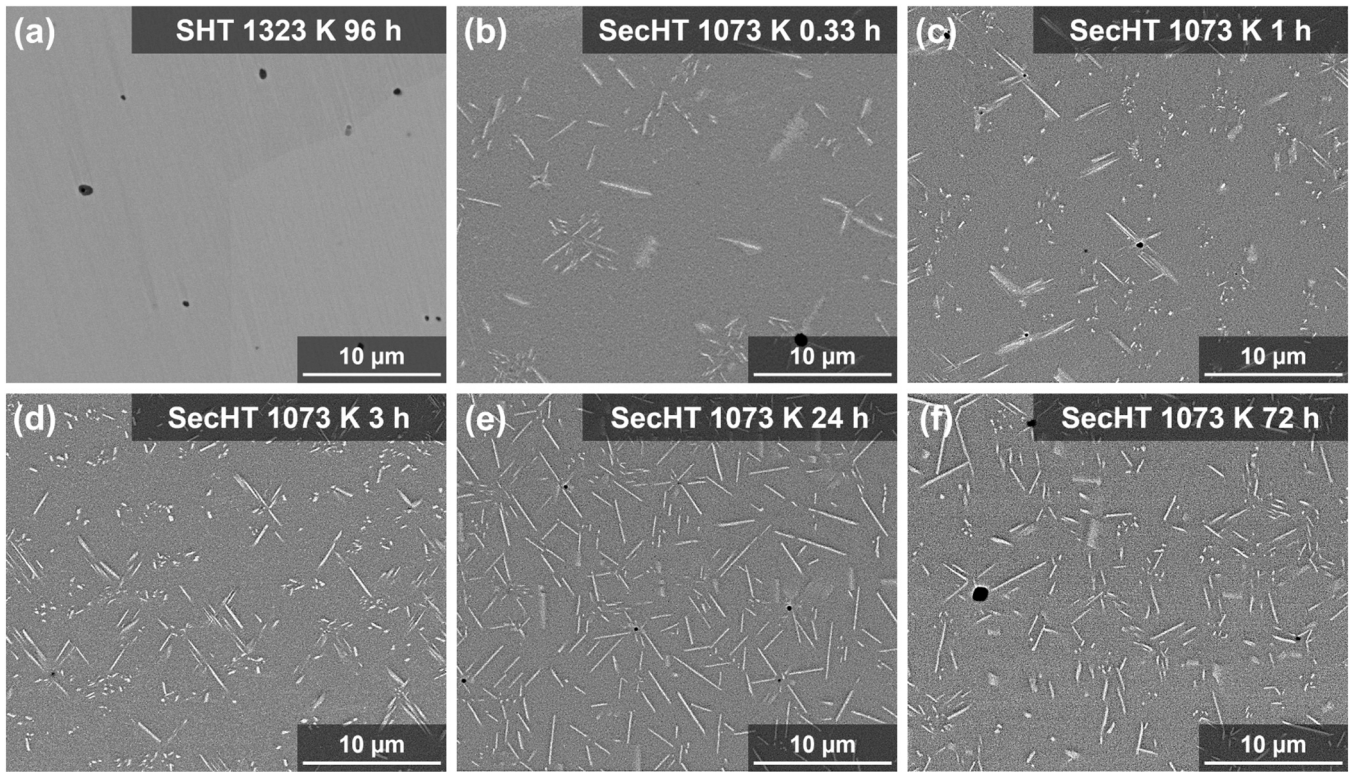


Fig. 6. Microstructures observed using Scanning Electron Microscopy (SEM) of $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ after high-SecHTs at 1073 K for various durations. (a) Homogenized condition (SHT) at 1323 K for 96 h. (b)-(h) Microstructures after SHT followed by SecHTs at 1073 K for different durations.

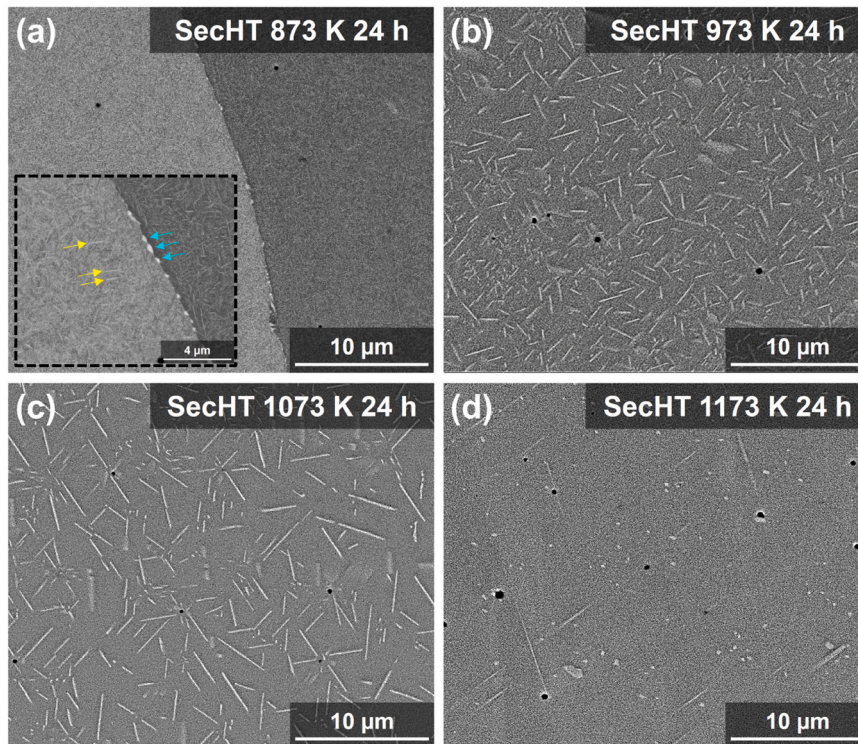


Fig. 7. Microstructures observed using Scanning Electron Microscopy (SEM) of $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ after high-temperature SecHTs at (a) 873 K/24 h, (b) 973 K/24 h, (c) 1073 K/24 h, and (d) 1173 K/24 h. Inset in (a) demonstrates a magnified region of 873 K/24 h case where intragranular and intergranular precipitates are marked with yellow and cyan arrows, respectively.

size and, especially, volume fraction compared to the 1073 K/24 h case, likely due to increased solubility of the second phase in the matrix at

1173 K.

Precipitates are observed both within grains (intragranular) and

along grain boundaries (intergranular) for all SecHT conditions. Intragranular precipitates, which are the most common, exhibit a sharp, needle-like morphology, while intergranular ones appear bulkier and rounder, as seen in the inset of Fig. 7(a).

Fig. 8 presents chemical distribution maps for a Co13.5 sample subjected to 1073 K/24 h SecHT, which contains multiple precipitates as revealed by the SEM-BSE micrograph in the inset of Fig. 8(a). These maps point out that the precipitates are clearly Ni-rich, Co-lean, Mn-lean, and Ti-rich compared to both the matrix and the nominal alloy composition. Notably, Ni and Mn species present the largest precipitate-matrix composition difference while Co shows the mildest. Later, it will be shown after EDS measurements on TEM thin foils that precipitate-matrix composition difference is larger than evidenced by Fig. 8. This difference is due to the very thin shape of the precipitates, for which the lower interaction volume, and thus higher spatial resolution, of measurements on the TEM samples results in more accurate determination of precipitate composition.

Fig. 9 presents room temperature TEM bright field images of a Co14.5 sample in the SHT state (1373 K for 96 h) (Fig. 9(a)), and Co13.5 samples in the SHT condition (Fig. 9(b)), after additional SecHT at 1173 K for 1 h (Fig. 9(c)), and at 873 K for 24 h (Fig. 9(d)). In good agreement with SEM observations, both compositions exhibit microstructures free of precipitates in the SHT state (Figs. 9(a) and 9(b)), but they show different phases. The bright field of the Co14.5 sample reveals a clean microstructure, later confirmed to be austenite using electron diffraction patterns (as shown in Fig. 10). In contrast, the Co13.5 sample displays a hierarchical tweed martensitic microstructure, as it transforms above room temperature.

The SecHTed Co13.5 samples (Figs. 9(c) and 9(d)) exhibit microstructures with distinctly different precipitate distributions. In particular, the 1173 K/1 h sample shows a low density of large needle-like micron-sized precipitates, while the 873 K/24 h sample displays a much higher density of similar needle-like precipitates, though smaller in size and with reduced interparticle spacing on average. In addition to the differing precipitate distributions, the 1173 K/1 h and 873 K/24 h also

exhibit different matrix phases: martensite and austenite, respectively. This change in the matrix structure is consistent with the results in Fig. 4, as the latter thermal processing condition decreases the MT critical temperatures well below room temperature.

Fig. 10(a) shows a selected area electron diffraction pattern (SAEDP) captured for a Co14.5 SHT sample along the [110] zone axis of the austenite. This sample was selected because the higher Co content compared to the corresponding Co13.5 samples reduces MT temperatures (see Fig. 3(a) and Table 1), making it suitable for observing the austenite phase isolated from martensite and precipitate phases. The electron diffraction pattern clearly shows the typical fundamental spots of the BCC structure, coupled with additional reflections attributed to the B2 ordered phase, such as (001) or $\bar{1}11$, as indicated in Fig. 10(a). No evidence for the additional spots indicating the presence of L2₁ ordering was observed. The presence of B2 features without L2₁ ordering is consistent with the previous studies on Ni-(Co)-Mn-Ti alloys [28,43,63,64].

The high Ni and Ti contents observed in the precipitates (Fig. 8) suggests potential similarities between the secondary phases formed in the Ni-Co-Mn-Ti system and those in conventional Ni-rich Ni-Ti alloys. In Ni-Ti alloys, various precipitate phases, such as Ni₃Ti, Ni₃Ti₂, and Ni₄Ti₃, form under different thermal conditions, depending on temperature and duration of heat treatments [65,66]. Consequently, it would be valuable to investigate whether the precipitates observed in the present Ni-Co-Mn-Ti alloys exhibit structural or compositional similarities to those in the Ni-Ti system.

Fig. 10(b) displays a SAEDP obtained along the [111]_{B2} zone axis in a region of the Co13.5 sample after 24 h at 873 K, containing both the austenitic matrix and a high density of precipitates, as shown in Fig. 9(d). The intense diffraction spots correspond to the austenite matrix, while the additional spots along the three equivalent $\langle 01\bar{1} \rangle_{B2}$ directions of the austenite can be indexed to a hexagonal Ni₃Ti-type structure (space group $P6_3/mmc$). The interplanar distances along these directions do not match those of the Ni₃Ti₂ or Ni₄Ti₃ phases, effectively ruling out their presence in this alloy system. Additionally, spots

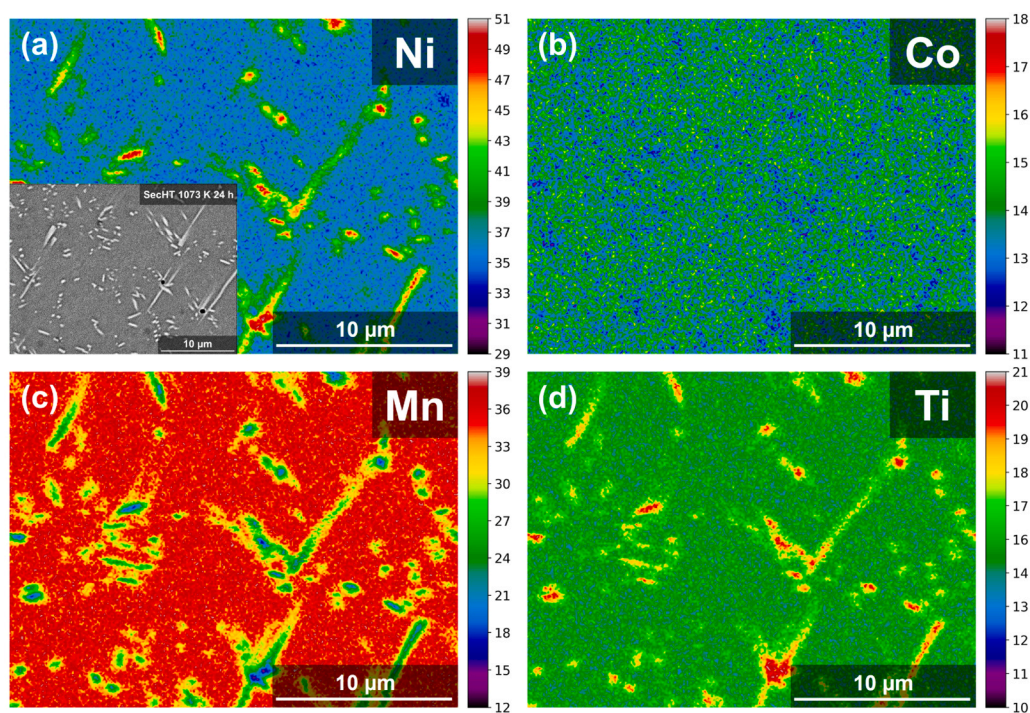


Fig. 8. Wavelength dispersive X-ray spectroscopy (WDS) mapping of Ni_{36.5}Co_{13.5}Mn₃₅Ti₁₅ after SecHT at 1073 K for 24 h. Panels show the spatial distribution of (a) Ni, (b) Co, (c) Mn, and (d) Ti (color scales in at.%). The inset in (a) is a secondary-electron SEM micrograph of the same region, highlighting the needle-like precipitates. Composition results confirm that the precipitates are Ni-/Ti-rich and Mn-/Co-lean.

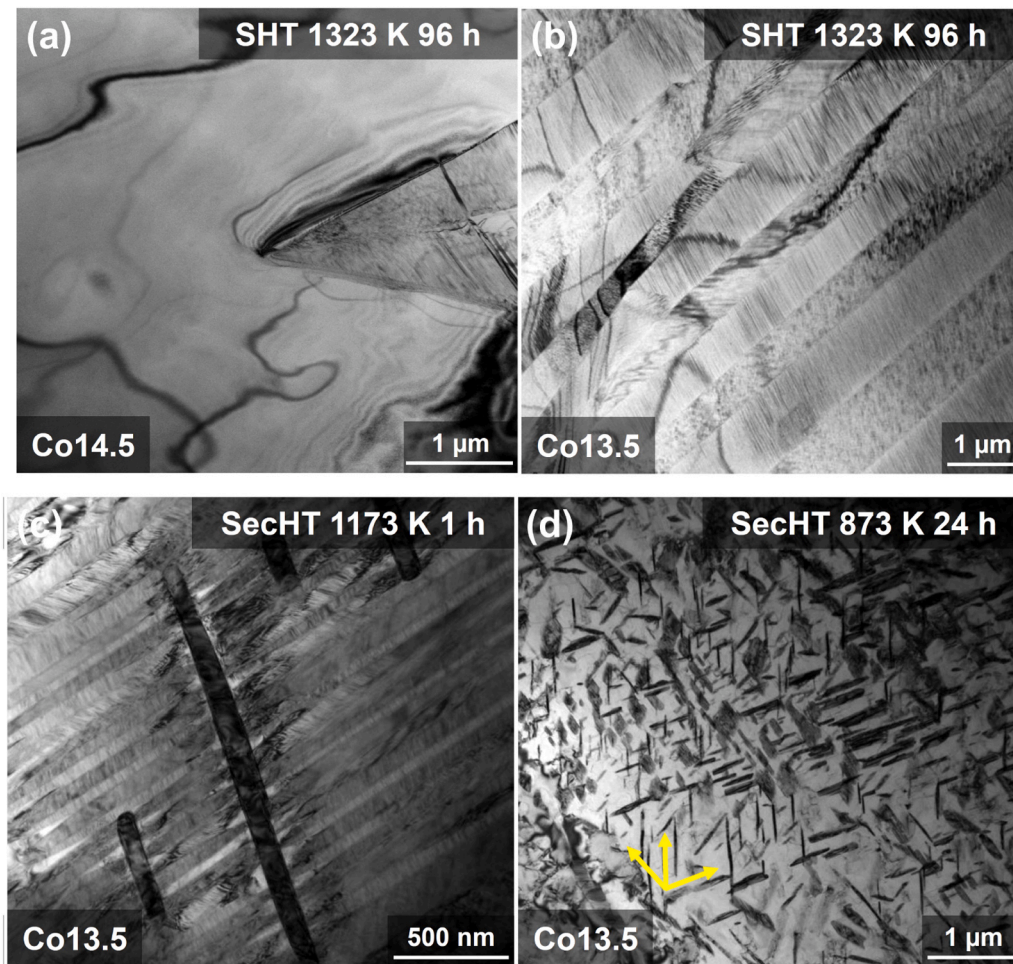


Fig. 9. Transmission electron microscopy (TEM) bright field images of (a) a $\text{Ni}_{35.5}\text{Co}_{14.5}\text{Mn}_{35}\text{Ti}_{15}$ sample in SHT state and $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ samples with (b) only SHT or with SecHTs at (c) 1173 K for 1 h or (d) at 873 K for 24 h. Yellow arrows in (d) are aligned with the long axis of the precipitates, indicating the general orientation of each precipitate variant. Note the different magnification scales.

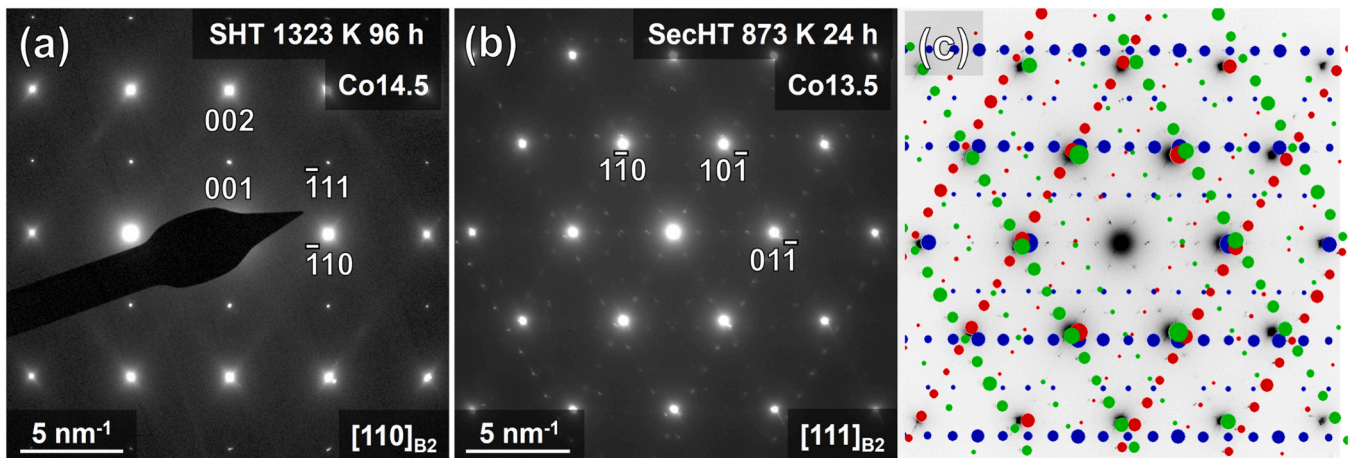


Fig. 10. SAEDP obtained for (a) a $\text{Ni}_{35.5}\text{Co}_{14.5}\text{Mn}_{35}\text{Ti}_{15}$ sample in SHT state (96 h at 1323 K) along the $[110]$ zone axis and (b) a $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ sample subjected to a SecHT at 873 K for 24 h along the $[111]$ crystallographic orientation of the austenitic matrix. Some spots corresponding to the cubic matrix are labeled. (c) Reverse grayscale version of the diffraction pattern in (b), where spots corresponding to 3 different variants of the $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitates are indicated in blue, red and green.

originating from three different precipitate variants can be identified in the SAEDP, aligning with the three equivalent crystallographic directions of the austenitic matrix according to the expected epitaxial

relationship between the B2 matrix and the Ni_3Ti -type structure:

$$[111]_{\text{B2}} \parallel [2\bar{1}10]_{\text{Ni}_3\text{Ti}} (01\bar{1})_{\text{B2}} \parallel (0004)_{\text{Ni}_3\text{Ti}}$$

Indeed, bright field images obtained close to $[111]_{\text{B2}}$ -type zone axes

reveal that the ellipsoidal precipitates predominantly orient their long axes along three distinctive directions (yellow arrows in Fig. 9(d)). To better illustrate this diffraction pattern, Fig. 10(c) shows the SAEDP (inverse contrast) from Fig. 10(b), together with simulated diffraction patterns under kinematical conditions for three Ni_3Ti -type precipitate variants, differentiated by color. Each variant aligns its $[0001]^*$ direction with one of the $\langle 01\bar{1} \rangle_{\text{B2}}$ directions in the matrix, as previously stated. Although prior studies on Ni-(Co)-Mn-(Fe)-Ti alloys have reported the formation of Ni_3Ti -type precipitates after SHT [30,67–69] and SecHT [54], none have identified this specific epitaxial relationship with the B2 matrix. Finally, it is noted that a few additional spots not predicted by the simulated pattern appear in Fig. 10(c). These spots can be attributed to double diffraction phenomena, which are not accounted for under kinematic conditions [70].

Energy-dispersive X-ray spectroscopy (EDS) performed on various Co13.5 samples confirms the precipitate composition as approximately $\text{Ni}_{64.4}\text{Co}_{10.8}\text{Mn}_{2.2}\text{Ti}_{22.6}$ (at.%), closely matching the formula $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$. This suggests that Ni and Co predominantly occupy Ni sites in the Ni_3Ti precipitates of the binary alloy, while Ti (and potentially Mn) occupy the remaining positions. In summary, all Co13.5 samples subjected to secondary heat treatments between 873 K and 1173 K consistently exhibit Ni_3Ti -type precipitates with identical structural and compositional characteristics. This confirms that the Ni-Co-Mn-Ti alloy system does not form other Ni-rich Ni-Ti family precipitates, at least within the range of aging durations and temperatures explored in this work.

Additional evidence regarding the nature of $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitates was collected using XRD. Fig. 11 shows the X-ray diffraction pattern of a Co13.5 SHT sample measured at 373 K, alongside samples subjected to SecHT at 973 K for 3 h, 24 h, and 72 h, measured at RT. These SecHT conditions were selected because their MT temperatures lie below RT, and they exhibit a relatively high precipitate area fraction confirmed by image analysis. This allows enhanced diffracted intensity from precipitates while avoiding overlapping peaks associated with the martensitic phases. Black squares indicate the main peaks of the austenite, (110), (200), and (211), which are the only peaks observed in the precipitation-free SHT sample. Black circles indicate peaks consistent with D0_{24} hexagonal crystalline structure expected for the $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitates, ($2\bar{2}01$), ($2\bar{2}02$), ($2\bar{2}03$) and ($2\bar{2}40$), as observed in samples subjected to SecHT at 973 K. These results align well with the diffraction patterns observed via TEM for the 1173 K/1 h and 873 K/24 h conditions.

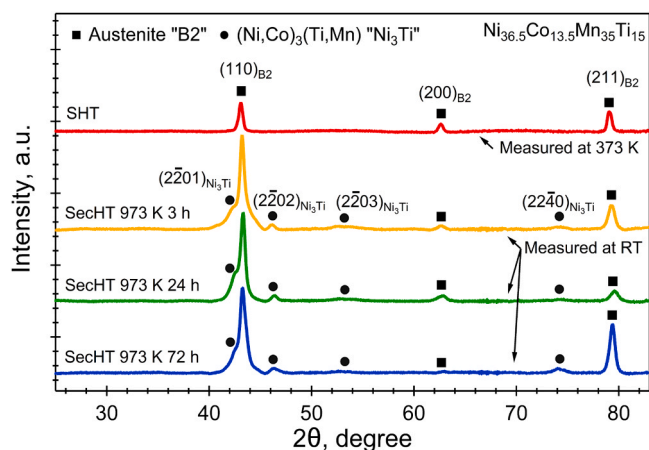


Fig. 11. XRD results of $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ SHT measured at 373 K and after SecHTs at 973 K for 3 h, 24 h, and 72 h, measured at room temperature. Black circles indicate peaks related to precipitates with D0_{24} hexagonal crystal structure, while black squares are related to the cubic (B2) austenite matrix.

4. Discussion of the results

Figs. 4 and 5 clearly demonstrate that sufficiently long SecHTs at high temperatures significantly influence the MT characteristics of Ni-Co-Mn-Ti alloys. Specifically, these treatments alter transformation temperatures, thermal hysteresis (A_F-M_S), and magnetization changes associated with the MT (ΔM). Microstructural analysis from SEM, TEM, and XRD provides strong evidence linking these alterations to the formation and evolution of $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitates. These precipitates affect the MT primarily through compositional changes in the surrounding matrix, modifying the matrix chemistry, particularly Ni and Mn concentrations, which impact the transformation temperatures and magnetic properties. Additionally, potential secondary factors such as changes in atomic ordering and precipitate-induced mechanical effects, including local hardening or nucleation constraints, may further influence MT behaviors. In the following subsections, these contributions are systematically examined to elucidate the microstructural and compositional mechanisms underlying the sensitivity of MT properties to thermal processing in the Ni-Co-Mn-Ti alloy system.

4.1. Precipitate formation and growth by secondary heat treatment

Figs. 6 and 7 show that present Ni-Co-Mn-Ti alloys form large concentrations of $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitates within the same range of thermal processing conditions that cause significant changes in MT characteristics, making such precipitation the primary candidate responsible for the observed evolution of the MT behavior. Supporting this, Fig. 12 demonstrates a clear negative correlation between the M_S temperature and precipitate area fraction, as measured from the SEM-BSE images using image analysis software. In general, lower SecHT temperatures and longer durations result in higher precipitate area fractions and consequently lower M_S temperatures.

Given the chemical difference between the precipitates, measured via TEM EDS as $\text{Ni}_{64.4}\text{Co}_{10.8}\text{Mn}_{2.2}\text{Ti}_{22.6}$, and the nominal composition of the alloy, $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$, this precipitation phenomenon should significantly impact on the matrix composition undergoing the MT, as detailed in Fig. 13 for Ni and Mn atomic species, and anticipated from Fig. 8. These EDS measurements reveal a clear decrease in Ni and an increase in Mn concentrations within the matrix as the precipitate area fraction increases. According to the measured precipitate composition, the formation of 7% volume fraction of precipitates is expected to change the matrix composition approximately by -2.1 , $+0.2$, $+2.4$ and -0.6 at.% for Ni, Co, Mn, and Ti atomic species, respectively. This prediction aligns well with the observed trends shown in Fig. 13,

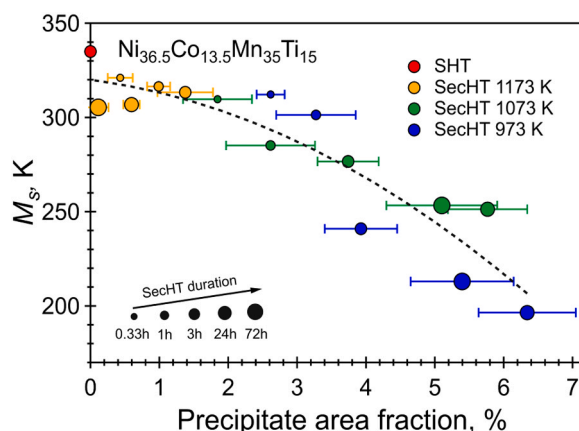


Fig. 12. Martensite start temperature, M_S , as a function of precipitate area fraction in $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ samples subjected to SHT only (red) or followed by SecHTs performed at 973 K (blue), 1073 K (green), or 1173 K (orange) for various durations. Marker size corresponds to the duration of the SecHT.

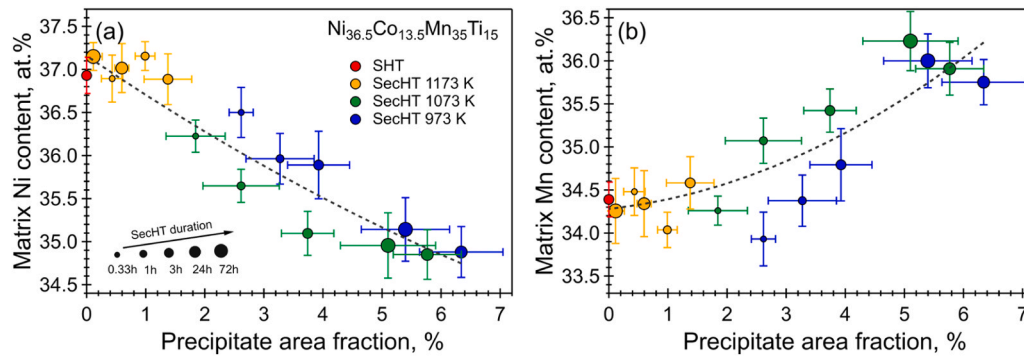


Fig. 13. Matrix concentrations of (a) Ni and (b) Mn as a function of precipitate area fraction in $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ samples after SecHTs at 973 K (blue), 1073 K (green), and 1173 K (orange). Marker size scales with the duration of each SecHT.

confirming that the primary compositional changes occur in Ni and Mn within the matrix.

These changes in matrix composition clearly correlate with the variation in MT temperatures, as shown in Fig. 14 for M_S . Similar evolution of M_S due to the substitution of Ni by Mn atoms has been previously observed [71,72], where this correlation was attributed to a decrease in the valence electron per atom ratio (e/a). Indeed, in the present Co13.5 alloy, the e/a ratio changes from 8.08 for the SHT condition to 8.01 for the 973 K/24 h condition. For comparison, previous works have shown that a compositional change in the ternary $\text{Ni}_{50}\text{Mn}_{50-x}\text{Ti}_x$ system can produce an M_S decrease of approximately 1275 K per unit change in e/a [73], corresponding to about a 90 K reduction for the compositional shift observed in Fig. 13. This estimation is likely a lower bound since it does not account for the additional stabilization of magnetic austenite relative to martensite as MT temperatures decrease below T_C , enhancing the magnetic entropy difference associated with the MT [59,60]. Therefore, changes in e/a ratio likely contribute significantly to the observed reduction in MT temperatures shown in Fig. 14. However, it is important to note that most studies examining the effects of the e/a ratio focus on pseudo-stoichiometric $\text{Ni}_{50}\text{Mn}_Y\text{X}_Z$ compositions, where $Y + Z = 50$ at. %.

This pseudo-stoichiometric relationship is, however, not conserved during the present precipitation process. Assuming that the two atomic sites of the B2 structure are preferentially occupied either by Ni and Co (Ni sites) or by Mn and Ti (Mn sites), as observed in other Ni–Mn-based MMSMs [22,74], the precipitation of the $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ second phases results in the matrix becoming distinctly “Ni-lean”. Following this consideration, Fig. 15 presents the evolution of the MT temperatures and the T_C of the austenite as a function of $(\text{Ni}+\text{Co})/(\text{Mn}+\text{Ti})$ ratio. This clearly demonstrates that most of the thermal processing effects on the M_S temperature occur when this elemental ratio is below 1, indicating a

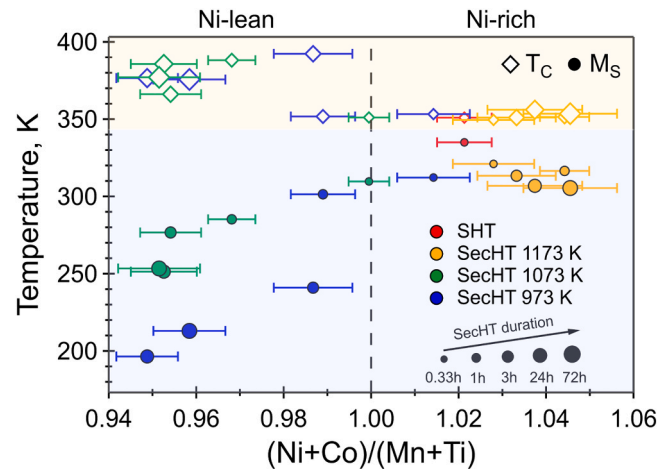


Fig. 15. Dependence of Curie temperature (T_C , open diamonds) and martensite start temperature (M_S , filled circles) on the atomic ratio $(\text{Ni}+\text{Co})/(\text{Mn}+\text{Ti})$ for Ni-lean (ratio < 1) and Ni-rich (ratio > 1) compositions. The vertical dashed line at $(\text{Ni}+\text{Co})/(\text{Mn}+\text{Ti}) = 1$ indicates the boundary between Ni-lean and Ni-rich composition regimes. Blue and yellow background colors remark an arbitrary boundary to facilitate the distinction between the data for M_S and T_C temperatures, respectively.

“Ni-lean” matrix. Additionally, T_C is higher for the “Ni-lean” samples than for the “Ni-rich” samples. A possible explanation for this difference is that excess Mn atoms must occupy Ni/Co sites, thereby becoming nearest neighbors with Mn atoms in Mn/Ti sites. These newly formed Mn–Mn pairs would predominantly exhibit a positive magnetic exchange

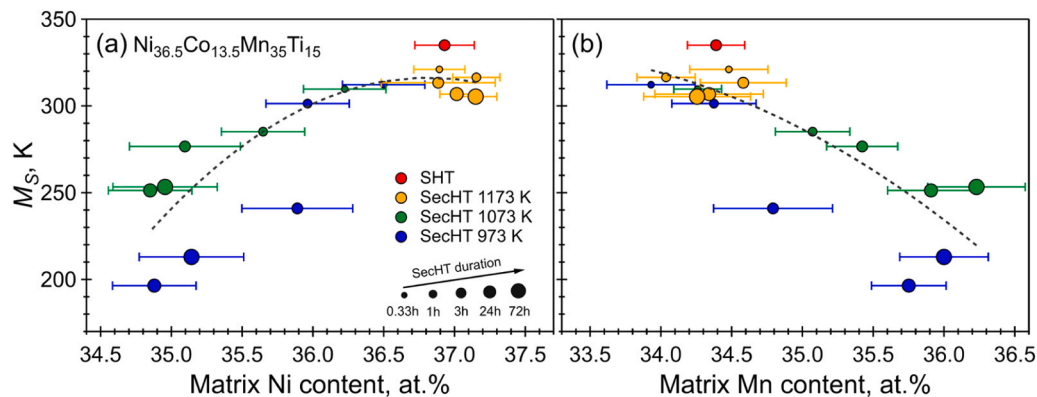


Fig. 14. Martensite start temperature, M_S , plotted against matrix content of (a) Ni and (b) Mn content in $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$ samples after SecHTs at 973 K (blue), 1073 K (green), and 1173 K (orange). Marker size corresponds to the duration of each treatment.

interaction [13], raising the T_C and stabilizing the ferromagnetic austenite, thus lowering M_S . Furthermore, because the precipitates are Co-lean relative to the nominal composition, one might anticipate a slight increase in the matrix Co content, potentially enhancing ferromagnetic interactions; however, no such variation was detected by any EDS measurements conducted in this study, suggesting this contribution is negligible.

These compositional trends are consistent with the recent study by Khan et al. [72], who systematically varied the Mn/Ni ratio in a series of $\text{Ni}_{50-x}\text{Co}_{15}\text{Mn}_{35+x}\text{Ti}_{15}$ Heusler alloys. All samples in that study were subjected to the same SHT (1273 K for 4 days), so composition is the primary variable across the series. The resulting compositional changes follow the same trend observed in the present study during precipitation, namely, decreasing Ni, increasing Mn, and slightly decreasing Ti in the matrix. When expressed in terms of the $(\text{Ni}+\text{Co})/(\text{Mn}+\text{Ti})$ ratio, their alloys span a range 0.996–0.948, which overlaps well with the SecHT-modified matrix compositions in this work (1.014–0.949). Over this common compositional range, the martensite start temperature (M_S) in the present study decreases from approximately 300 K to 196 K, while Khan et al. report a decrease from approximately 282 K to 120 K. Both datasets, therefore, show a large and consistent reduction in M_S with decreasing $(\text{Ni}+\text{Co})/(\text{Mn}+\text{Ti})$ ratio. The systematic offset in M_S between the two studies at comparable ratios is likely due to the difference in Co content (approximately 15 versus 13.5 at.%), as higher Co content is known to strengthen ferromagnetic ordering in the austenite phase, further stabilizing it relative to the martensite and reducing M_S .

Samples treated at 873 K are expected to follow the same trends correlating precipitate area fraction, matrix chemical composition, and transformation and magnetic properties discussed above. Nevertheless, the small size and high density of precipitates after SecHTs performed at 873 K hinder accurate quantification of the precipitate area fraction and matrix composition using the same methods as the other samples;

therefore, these samples are excluded from the analysis presented in Figs. 12–15. Acquiring data solely from the matrix was challenging, as measurements could unintentionally incorporate precipitate data, and quantifying precipitate area fractions was difficult due to the very small sizes of the precipitates.

4.2. Precipitation kinetics

The evolution of precipitate area fraction (PAF) with SecHT duration at 973 K, 1073 K, and 1173 K was fitted to the Johnson-Mehl-Avrami-Kolmogorov (JMAK) equation, $f(t) = f_{\max}(1 - \exp(-(kt)^n))$, where $f_{\max}$ is the equilibrium precipitate area fraction, k is the rate constant, and n is the Avrami exponent. The parameters $f_{\max}$, k , and n were obtained by weighted least-squares regression, using the measured standard deviation of PAF at each time point as the weighting factor. Data points showing a slight decrease in PAF (72 h at 973 K and 1073 K, and ≥ 24 h at 1173 K) were excluded, as a limited number of data points makes the fitted kinetic parameters highly sensitive to non-monotonic behavior. As shown in Fig. 16(a), the fitted JMAK curves capture the measured PAF evolution well at all three temperatures. The equilibrium fraction $f_{\max}$ decreases from 7.35% at 973 K to 6.38% at 1073 K and 1.80% at 1173 K, consistent with increasing solubility of $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitates at higher temperatures. The inset of Fig. 16(a) presents a linear extrapolation of $f_{\max}$ versus temperature, yielding an estimated precipitation solvus of ~ 1259 K. This estimate is consistent with the absence of precipitates after SHT at 1323 K and with the very low equilibrium fraction observed after SecHT at 1173 K. The extrapolated $f_{\max}$ at 873 K is $\sim 10.7\%$.

Fig. 16(b) shows the Arrhenius plot of $\ln(k)$ versus $1000/T$ for the three directly measured SecHT temperatures. The temperature dependence of the rate constant follows the Arrhenius relationship, $\ln(k) = \ln(k_0) - Q/(RT)$, where k_0 is the pre-exponential factor, Q is the

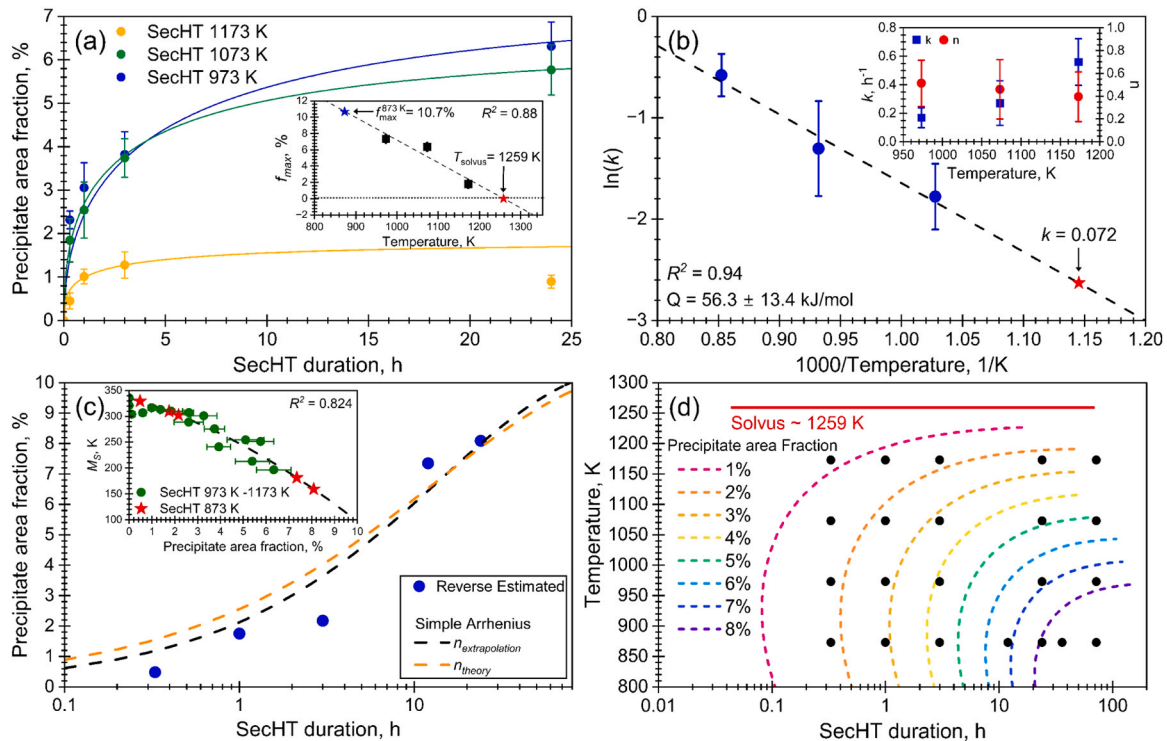


Fig. 16. Kinetic analysis of $(\text{Ni},\text{Co})_3(\text{Ti},\text{Mn})$ precipitation in $\text{Ni}_{36.5}\text{Co}_{13.5}\text{Mn}_{35}\text{Ti}_{15}$. (a) Precipitate area fraction (PAF) versus SecHT duration at 973 K (blue), 1073 K (green), and 1173 K (orange) with JMAK fits. Inset shows $f_{\max}$ versus temperature with linear extrapolation to the estimated solvus and to 873 K. (b) Arrhenius plot of $\ln(k)$ versus $1000/T$. Inset shows k and n values versus temperature. The red star marks the Arrhenius-extrapolated k at 873 K. (c) Reverse-estimated PAF at 873 K (blue circles) with the forward JMAK prediction (dashed line). The inset shows the M_S versus PAF quadratic fit used for the reverse estimation. (d) TTT diagram with iso-precipitate-fraction contours (1–8%) and experimental data points (black circles).

apparent activation energy, and R is the universal gas constant. The fit yields an apparent activation energy of $Q = 56.3 \pm 13.4$ kJ/mol. The inset of Fig. 16(b) displays the individual k and n values as a function of SecHT temperature. The rate constant k increases monotonically from 0.16 h^{-1} at 973 K to 0.27 h^{-1} at 1073 K and 0.56 h^{-1} at 1173 K, while the Avrami exponent n ranges from 0.395 to 0.513 across the three temperatures, with a mean value of $n_{\text{avg}} = 0.456$. Given the needle-like morphology of the D0_{24} precipitates (Figs. 6, 7, and 9), one-dimensional growth is expected ($N_{\text{dim}} = 1$). For diffusion-controlled growth ($g = 0$) with early nucleation site saturation ($B = 0$), the JMAK theory predicts $n_{\text{theory}} = N_{\text{dim}} \times g + B = 0.5$. The experimental values are very close to this prediction. Pandolfi et al. [75] reported similar Avrami exponents (0.32–0.53) for lenticular Ti_3Ni_4 precipitation in aged Ni-rich Ni–Ti alloys and attributed values slightly below 0.5 to heterogeneous nucleation. This interpretation is consistent with the present system, where precipitates nucleate both at grain boundaries and within grains (Fig. 7(a)).

The measured Q is notably lower than the 108 ± 8 kJ/mol reported by Pandolfi et al. for Ti_3Ni_4 precipitation in binary Ni–Ti, the 98 ± 12 kJ/mol reported by Shuitcev et al. [76] for H-phase in $\text{Ni}_{50}\text{Ti}_{30}\text{Hf}_{20}$, and the activation energy for bulk Ni diffusion in B2 NiTi (143 ± 11 kJ/mol) [77]. Shuitcev et al. demonstrated that the apparent activation energy for precipitation in Ni–Ti based SMAs can be significantly lower than the bulk diffusion activation energy and attributed this to the role of point defects in facilitating the precipitation process. In their system, $Q = 98$ kJ/mol matched the formation energy of Ti antisite defects on the Ni sublattice (93 kJ/mol) [78]. In the present Ni–Co–Mn–Ti system, a similar defect-assisted mechanism may operate. However, it should be noted that Q was extracted from only three temperatures spanning a 200 K range, giving a single degree of freedom, and the standard error of the Arrhenius slope yields an uncertainty of approximately ± 13.4 kJ/mol. A more precise determination of Q would require additional aging temperatures.

Direct measurement of PAF at 873 K by SEM image analysis is not feasible because the precipitates are too fine and densely distributed (Fig. 7(a) inset). To estimate the PAF at this temperature, an empirical correlation between M_s and PAF was established using all measured data at 973 K, 1073 K, and 1173 K. A constrained quadratic fit was used to invert the measured M_s values at 873 K and obtain the corresponding PAF estimates, as shown in the inset of Fig. 16(c). The resulting reverse-estimated PAF peaks at approximately 8.14% after 24 h of SecHT. Fig. 16(c) compares these reverse-estimated PAF values (blue circles) with the forward prediction from the Arrhenius-extrapolated rate constant ($k = 0.078 \text{ h}^{-1}$, dashed line), using the extrapolated $f_{\text{max}} = 10.7\%$ and two different n values. The first is $n_{\text{extrapolation}} = 0.575$, obtained by linear extrapolation of the Avrami exponents fitted at 973–1173 K to 873 K. The second is $n_{\text{theory}} = 0.5$, corresponding to the theoretical Avrami exponent for diffusion-controlled growth with early nucleation site saturation. The reverse-estimated PAF at 873 K/24 h is approximately 8.14%, which is the highest among all SecHT conditions and consistent with both Arrhenius-extrapolated JMAK predictions at this temperature. This result confirms that the 873 K condition, despite being inaccessible to direct SEM image analysis, produces the largest precipitate fraction and consequently the largest compositional shift in the matrix, fully accounting for the lowest M_s observed in Fig. 5(a).

Fig. 16(d) presents the time-temperature-transformation (TTT) diagram constructed by inverting the JMAK equation combined with the Arrhenius relationship to obtain the time required to reach a given PAF level at each temperature, $t = (1/k(T)) \times (-\ln(1 - f/f_{\text{max}}(T)))^{(1/n_{\text{avg}})}$ where $k(T)$ is obtained from the Arrhenius fit, $f_{\text{max}}(T)$ from the linear extrapolation shown in the inset of Fig. 16(a), and $n_{\text{avg}} = 0.456$, the average Avrami exponent from the three directly measured temperatures. The resulting C-shaped contours for PAF levels of 1% through 8% exhibit the characteristic nose near 950–1000 K, where the balance

between nucleation driving force and diffusion kinetics is optimal. At higher temperatures approaching the solvus, the contours shift to longer times due to the diminishing thermodynamic driving force. At lower temperatures, the contours also shift to longer times due to slower diffusion. The experimental data points at all four SecHT temperatures are overlaid and fall within the expected ranges, providing practical guidance for selecting SecHT conditions to achieve a desired precipitate area fraction in Ni–Co–Mn–Ti alloys.

4.3. Chemical/atomic ordering change by secondary heat treatments

Fig. 5(a) presents a clear dependence of MT temperatures on SecHT temperature, which can be explained by the increased volume fraction of precipitates and the resulting changes in matrix composition undergoing the MT. However, Figs. 12, 14, and 15 indicate that even for alloys with similar precipitate area fractions and compositions, there remains some dependency on SecHT temperature. Therefore, in Ni–Co–Mn–Ti alloys, additional factors influencing MT characteristics that are not related to compositional changes may exist.

Additionally, the contribution from potential changes in chemical (atomic) ordering during SecHTs should be considered, as such changes can significantly impact the magnetic and MT properties of Ni–(Co)–Mn–X MMSMAs [14,47,49]. Two key properties are particularly important for characterizing the influence of ordering transitions, such as B2–L2₁, on the MT of Ni–Mn–X alloys. The first property is the degree of atomic order, which can be reduced by defects such as anti-phase boundaries, anti-site defects, or quenched-in regions of higher disorder. Because ordering influences the distribution of Mn atoms within the crystal, and thus the number of Mn–Mn pairs interacting as first, second, or higher-order nearest-neighbors, it can modify the strength of ferromagnetic and antiferromagnetic interactions, affecting the magnetic entropy of the austenite and the thermodynamic balance between the austenite and martensite phases [79]. The second property is the correlation length scales of different features in the atomic order microstructure, such as ordered domain size or spacing between anti-phase boundaries. A correlation has been demonstrated between these length scales and the non-monotonic evolution of the MT in Ni–Co–Mn–In alloys, particularly when these length scales are below 100 nm, reaching a maximum effect around 30–40 nm [14].

A recent work by Koch et al. evaluated chemical long-range ordering in $\text{Ni}_{37}\text{Co}_{13}\text{Mn}_{33}\text{Ti}_{17}$ powder samples using neutron diffraction after annealing at either 973 K or 1173 K for 1 h [43]. The authors demonstrated that the 973 K/1 h sample exhibited a somewhat enhanced B2 degree of order compared to the 1173 K/1 h sample, and that varying SecHT temperature resulted in a decrease of M_s and an increase of T_c , similarly to the effect of an enhanced L2₁ ordering in Ni–Mn–Ga and Ni–Mn–In systems [58,59,80]. However, they also identified the formation of a low volume fraction of Ni_3Ti -like precipitates, whose diffraction peaks were more predominant on the 973 K/1 h sample than the 1173 K/1 h sample; the impact of these precipitates on matrix composition or MT properties was neither evaluated nor reported. Therefore, the work by Koch et al. strongly suggests that the B2 degree of order in Co13.5 samples should be affected by SecHTs between 873 K and 1173 K. Nonetheless, it remains challenging to precisely evaluate the impact of this change in the degree of order on MT properties due to the simultaneous precipitation phenomenon [43].

Although previous studies have shown that the impact of atomic degree of order on MT temperatures is mediated by changes in the ferromagnetic strength of the austenite [14], Fig. 15 demonstrates that the observed shift in M_s for samples annealed at 973 K and 1073 K occurs without a corresponding significant shift in T_c . In addition, previous studies suggest that the annealing duration required to achieve an equilibrium degree of order at relatively high temperatures is typically short, as only a limited number of atomic diffusion events are necessary to alter the degree of order (albeit with small correlation lengths) [48]. In other words, ordering is expected to occur much faster than

precipitate formation, provided the driving force is not extremely weak. For instance, in Ni–Co–Mn–In alloys heated to temperatures above 673 K, a thermal equilibrium between B2 and L2₁ phases (i.e., a constant order parameter) can be achieved within 15 min, sufficient to modify both T_C and M_S [14]. In contrast, Figs. 4 and 5 show that heat treatments between 873 and 1173 K for a duration of 20 min result in only minor changes in M_S and T_C .

Therefore, it is suggested that changes in the overall B2 degree of order in the present samples are highly likely during heat treatments performed between 873 and 1173 K in a Ni_{36.5}Co_{13.5}Mn₃₅Ti₁₅ alloy. However, the extent of their impact on MT properties remains unclear based on the evidence presented in this study, and the compositional comparison with Khan et al. [72], discussed in Section 4.1, indicates that any such impact is minor compared to the precipitation-induced compositional changes in the matrix. Neutron diffraction characterization is necessary to fully clarify this point and is proposed as a potential future work.

4.4. Mechanical effects

In SMAs, in addition to the chemical effects detailed above, precipitate formation can influence MT characteristics through mechanical interactions between precipitates and martensite nucleation [45,81,82]. The origin of these interactions can be attributed to two distinct mechanisms. On one hand, the mechanical effect arises from the constrained geometry of the matrix due to precipitate spacing and, thus, the increased energy required to form stable nuclei of reduced radius, as predicted by the classical nucleation theory. When the interparticle distance becomes comparable to or smaller than the critical radius required for martensite nucleation, greater free energy is needed to create stable nuclei, resulting in additional undercooling ($T_0 - M_S$), where T_0 is the thermodynamic equilibrium temperature between austenite and martensite, often assumed to lie halfway between M_S and A_f temperatures of thermoelastic MTs. Therefore, this effect should decrease M_S and widen thermal hysteresis [81,83]. On the other hand, thermoelastic martensitic transformation, being diffusionless and driven by displacive shear, is inherently sensitive to matrix hardening induced by precipitates.

For both mechanisms, precipitation size and volume fraction evolution are expected to result in non-monotonic changes in MT characteristics as a function of treatment duration, as a particular precipitate size and fraction can maximize mechanical effects by minimizing interparticle distances. For instance, Evirgen et al. observed this behavior in NiTiHf and NiTiZr alloys, heat-treated at relatively low aging temperatures (723 K and 773 K), which resulted in the formation of very fine H-phase precipitates, demonstrating that the maximum mechanical effect occurred at interparticle distances on the order of tens of nanometers [44–46]. Similarly, it has been shown that other microstructural features, such as atomic order domains in Ni–Mn–X MMSMAs can also facilitate this type of length-scale effect [14]. Length scales of the B2–L2₁ microstructure below 100 nm significantly affect MT properties of Ni₄₅Co₅Mn_{50-x}In_x alloys (with $x \approx 13.3$ –13.7 at.%), exhibiting a maximum mechanical effect around 30–40 nm. Similarly, Hornbogen correlated similar non-monotonic behavior of MT temperatures in Fe–Ni–Al alloys with yield strength (and thus, the microstructural length scales) [84]. Lin and Wu observed an inverse linear relationship between M_S and hardness in strain-hardened Ni–Ti alloys [85], which should again be due to the effects of microstructural length scales on martensite nucleation and growth.

Beyond interparticle spacing, an additional mechanical contribution may arise from the coherency of the precipitate–matrix interface. Coherent or semi-coherent precipitates impose elastic strain fields on the surrounding matrix that can stabilize the austenite phase and further suppress M_S . Chen et al. [86] demonstrated this in NiTiHfAl alloys. In that system, coherent Ni₂TiAl Heusler precipitates mechanically stabilized austenite, leading to a decrease in M_S . Upon loss of coherency, this

mechanical stabilization was removed and the chemical effect of Al depletion in the matrix dominated, resulting in an abrupt M_S increase of approximately 75 K. Whether a similar mechanism contributes to the partial M_S recovery observed between 24 and 72 h at 873 K and 973 K in the present study (Fig. 5(a)) remains an open question. The evolution of the precipitate–matrix interface and its role in the martensitic transformation should be investigated using high-resolution transmission electron microscopy in the future.

In this study, it is challenging to fully separate the mechanical effects of precipitates from their chemistry-driven effects, as both contribute to lowering MT temperatures. However, evidence suggests that mechanical contributions are minor compared to the dominant chemical effects. Precipitates formed under all SecHT conditions are in the fine- to coarse-micron size range, much larger than the nanometer-scale features previously shown to constrain martensite nucleation strongly. Additionally, early-duration SecHT at 873 K and 973 K, where precipitates are finest and most closely spaced, do not show greater reductions in M_S (Fig. 4) compared to higher-temperature SecHTs at 1073–1173 K. If mechanical constraints were significant, the largest M_S suppression would occur under these early-durations, low-temperature conditions. Instead, the initial M_S evolution with SecHT duration is consistent across all temperatures, with significant differences only appearing at long durations when matrix composition has clearly changed. The only exception is the modest M_S recovery of approximately 17–20 K observed between 24 and 72 h at 873 K and 973 K, which may be related to the coherency mechanism discussed above.

These microstructural and thermomagnetic trends suggest that changes in matrix composition due to precipitation are the primary factor influencing the observed SecHT-dependent MT behavior. Mechanical effects from micron-scale precipitates, if present, play only a secondary role. Other mechanisms, such as ordering-related nanoscale domains that could impose mechanical constraints, also appear unlikely. TEM dark-field imaging of a Co14.5 SHT sample along the (100) B2 direction showed no contrast for nanoscale ordered domains, suggesting the B2 phase is relatively homogeneous. It does not form features that impede martensite nucleation. Overall, the evidence suggests that precipitation-induced chemical change in the matrix governs the shifts in M_S , while mechanical nucleation barriers contribute minimally under the present SecHT conditions.

5. Summary and conclusions

In this study, it has been shown that systematic thermal processing at temperatures between 873 K and 1173 K provides precise control over the microstructure and martensitic transformation of Ni_{36.5}Co_{13.5}Mn₃₅Ti₁₅ alloys. Temperature-dependent magnetization measurements reveal that the martensitic start temperature (M_S) and the Curie temperature (T_C) depend on both temperature and duration of secondary heat treatments (SecHT). Specifically, SecHT at 873–1073 K generally decreases M_S and increases T_C , and these changes become more pronounced with extended durations. This behavior reflects the coupled influence of precipitation on structural and magnetic transitions. At the same time, higher-temperature treatments at 1173 K promote partial dissolution of precipitates, limiting further variations in both M_S and T_C .

A detailed SEM analysis shows the formation and growth of needle-shaped precipitates during SecHTs. These precipitates coarsen in both length and width as treatment time increases. At lower temperatures (873–973 K), slower diffusion kinetics produce finer, more evenly dispersed precipitates, whereas at 1173 K, partial solvus dissolution results in fewer, but much larger, precipitates. TEM imaging and SAEDPs further confirm that these precipitates are (Ni,Co)₃(Ti,Mn) with a D0₂₄ hexagonal structure. The SAEDPs reveal diffraction spots corresponding to three precipitate variants, each aligned with crystallographically equivalent orientations of the austenitic matrix, consistent with the expected epitaxial relationship between the B2 matrix and Ni₃Ti-type precipitates.

Compositional analyses using EDS elemental maps clearly show that as the precipitate area fraction increases, the surrounding B2 matrix correspondingly becomes significantly Mn-rich and Ni-lean. Quantitative correlations reveal a clear relationship between precipitate area fraction, matrix composition, and MT temperatures, identifying the chemical effects of precipitation on matrix composition as the main mechanism controlling the MT properties after SecHTs. It is suggested that these matrix compositional changes directly impact the martensitic transformation by altering the valence electron concentration per atom and increasing Mn-antisite defects within the B2 crystal structure. Other changes in the degree of B2 atomic order not mediated by compositional changes, i.e. development of the order-disorder reaction, are considered plausible during the SecHTs conducted in this study. However, no effect attributable to such changes was observed on the martensitic transformation or, especially, on the ferromagnetic transition of the austenite, as would be expected from the behavior of similar Ni–Mn-based MMSMAs.

Ultimately, this study delineates the interplay among thermal processing parameters, microstructural evolution, and the resulting magnetostructural properties in Ni–Co–Mn–Ti metamagnetic shape memory alloys. It thus provides strategies for controlling and optimizing functional performance through targeted microstructural control that, in turn, tunes the matrix chemistry at fixed nominal compositions.

CRedit authorship contribution statement

Woohyun Cho: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Ibrahim Karaman:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Jaume Pons:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Ruben Santamarta:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Daniel Salas:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Serdar Torun:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ibrahim Karaman reports financial support was provided by U.S. Army Research Office. Ruben Santamarta reports financial support was provided by Spain Ministry of Science Innovation and Universities. Jaume Pons reports financial support was provided by European Regional Development Fund. Ibrahim Karaman reports financial support was provided by US Department of the Army. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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